Complete figure book: clinical course, serum markers and EASI estimation

This review copy contains the five main figures followed by all 21 supplementary figures. Captions specify populations, uncertainty and what each comparison can establish. Complete numerical results are supplied as CSV source tables.

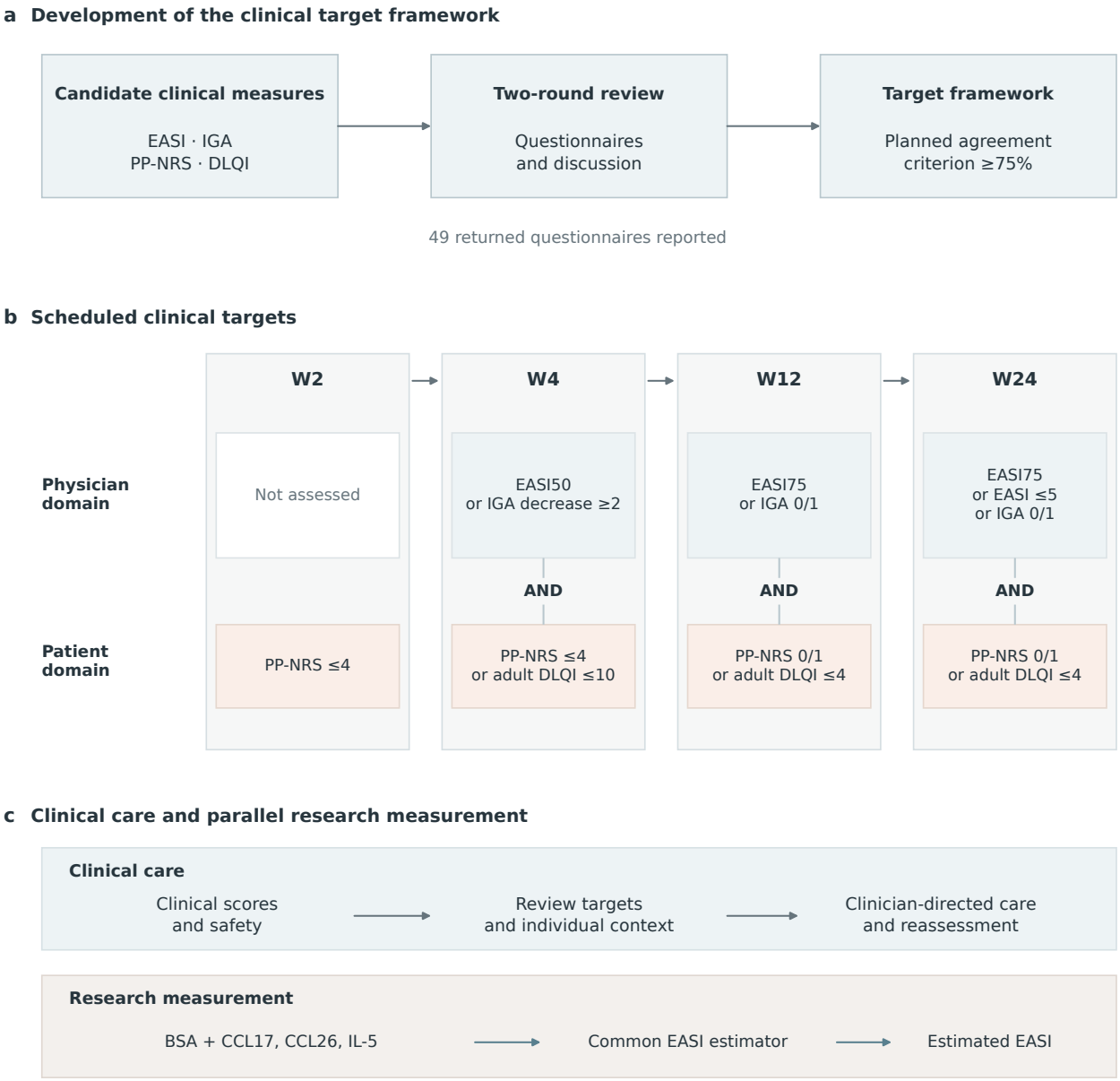


Figure 1: **Clinical target framework and parallel research EASI estimation.** a, Framework development described in the investigator materials: candidate clinical measures, a two-round questionnaire and discussion procedure, 49 returned questionnaires, and a planned agreement criterion of at least 75%. Original ballots and round-specific agreement were unavailable; these are source-reported development details, not independently verified consensus results. b, Scheduled targets transcribed from the project framework. Within each domain, any listed criterion is sufficient; from W4 onward both domains must be met. W2 assesses only PP-NRS $\leq$4. EASI50 and EASI75 denote at least 50% and 75% improvement from a positive observed baseline EASI. IGA decrease $\geq$2 is relative to baseline; IGA 0/1 and PP-NRS 0/1 denote observed values 0 or 1. DLQI targets apply to adults aged $\geq$18 years. Missing components are handled with three-valued logic, rather than automatically classified as target failure. The exact IGA instrument and approved implemented framework version require investigator confirmation. Visits are schematic and not spaced in proportion to time. c, Clinician-directed assessment and care are separated from the research estimator using BSA, CCL17, CCL26 and IL-5. The estimator was not a dose-change rule, and this observational analysis does not establish the effectiveness of the proposed integrated management strategy.

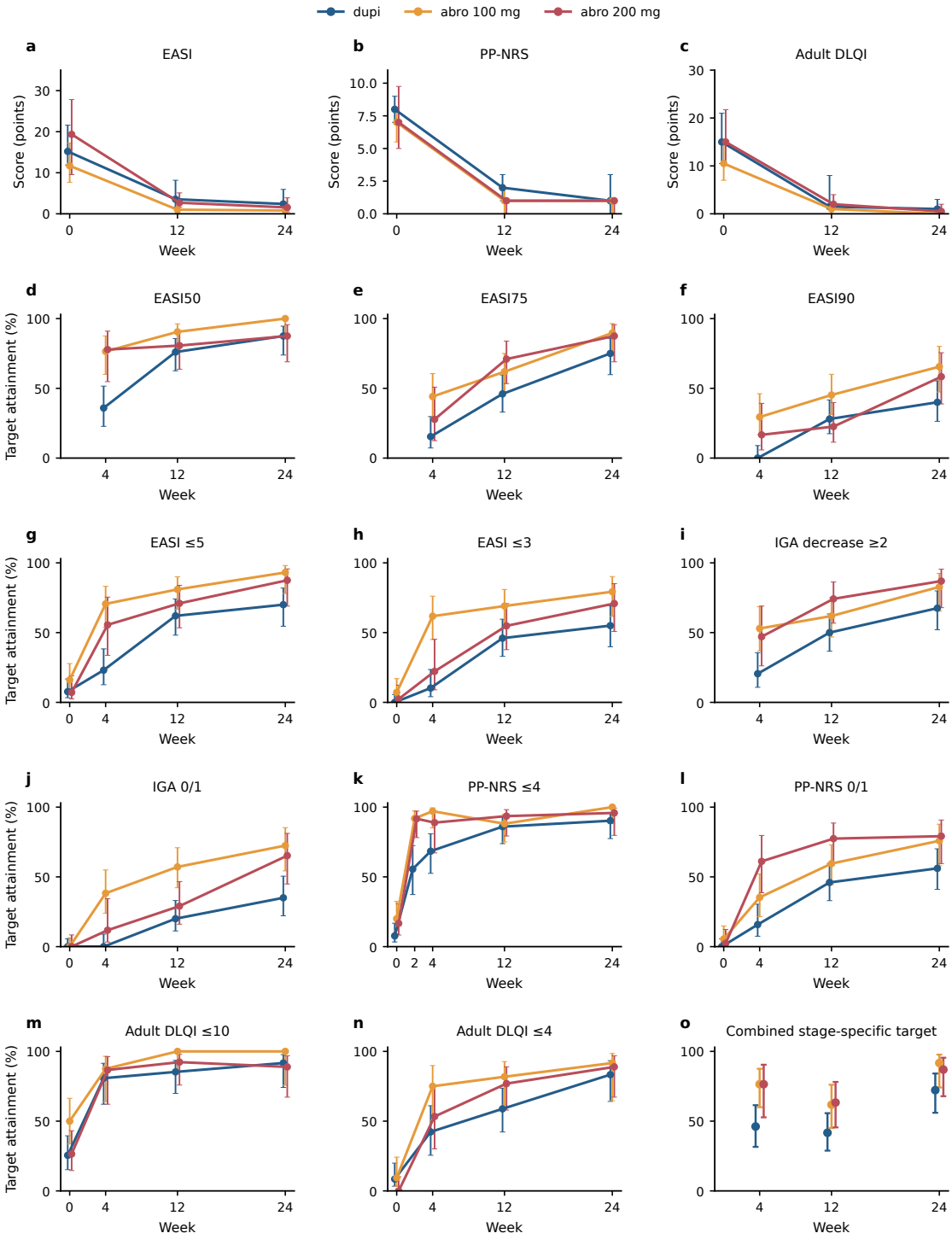


Figure 2: **Observed clinical activity and target attainment by initial treatment.** a–c, Observed medians and interquartile ranges for EASI, PP-NRS and adult DLQI. d–n, Clinical threshold attainment, with pointwise Wilson 95% confidence intervals. Each visit uses its available observations, not a fixed patient set. Group positions are slightly offset for legibility. Improvement thresholds require observed baseline and follow-up; adult DLQI requires age ≥18. o, Combined physician-and-patient targets from Fig. 1, with Wilson intervals. These stage-specific criteria differ, so points are not connected. Colour denotes initial treatment throughout; subsequent treatment changes do not reassign groups. No outcomes are imputed. Counts and missing-outcome bounds are in Supplementary Table S1 and source data; baseline-adjusted treatment comparisons and additional designated visits are in the supplement. These available-case, observational trajectories do not establish causal treatment differences.

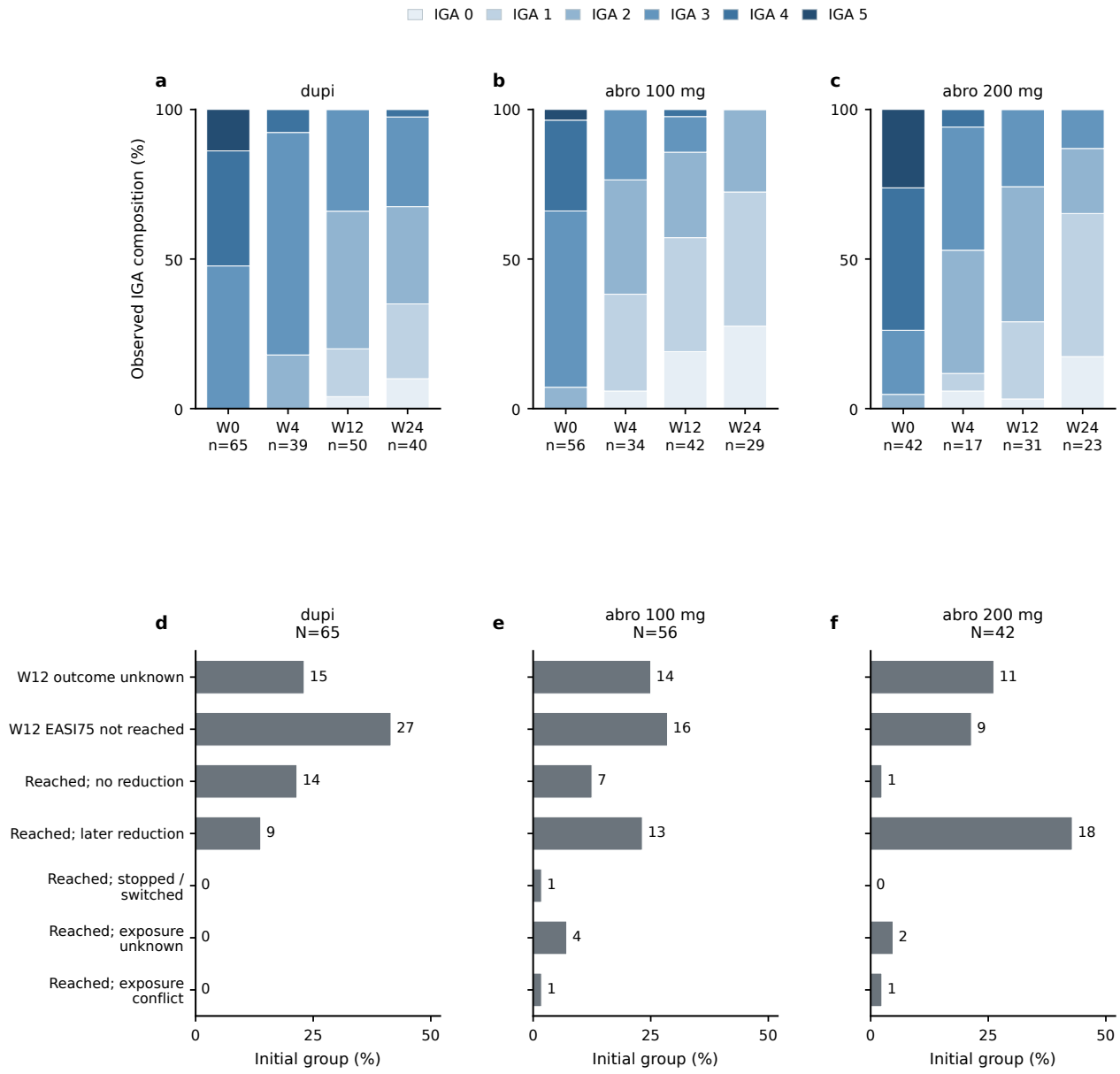


Figure 3: **Recorded skin severity and treatment-reduction pathways.** a–c, Recorded IGA compositions at W0, W4, W12 and W24. Each bar totals 100% of the evaluable patients; n below the bar is its patient denominator. The ordered blue scale identifies recorded IGA grades 0–5 and is not a treatment colour. No score or count is superimposed on the segments. All grade-specific counts and proportions remain in source tables. d–f, Future-defined pathways formed by W12 EASI75 status and subsequent recorded treatment changes. Horizontal bar length is the proportion of the complete initial group (N=65/56/42), and the number at its end is the pathway size. The seven mutually exclusive rows exhaust each initial group, including zero-size and uncertain-exposure rows. Reached refers to W12 EASI75; later reduction means a documented reduction after W12. These are descriptive pathways, not randomized reduction strategies; the bars do not encode subsequent efficacy or relapse. Instrument identity remains unverified for the recorded IGA scale. Complete pathway counts are available in source data.

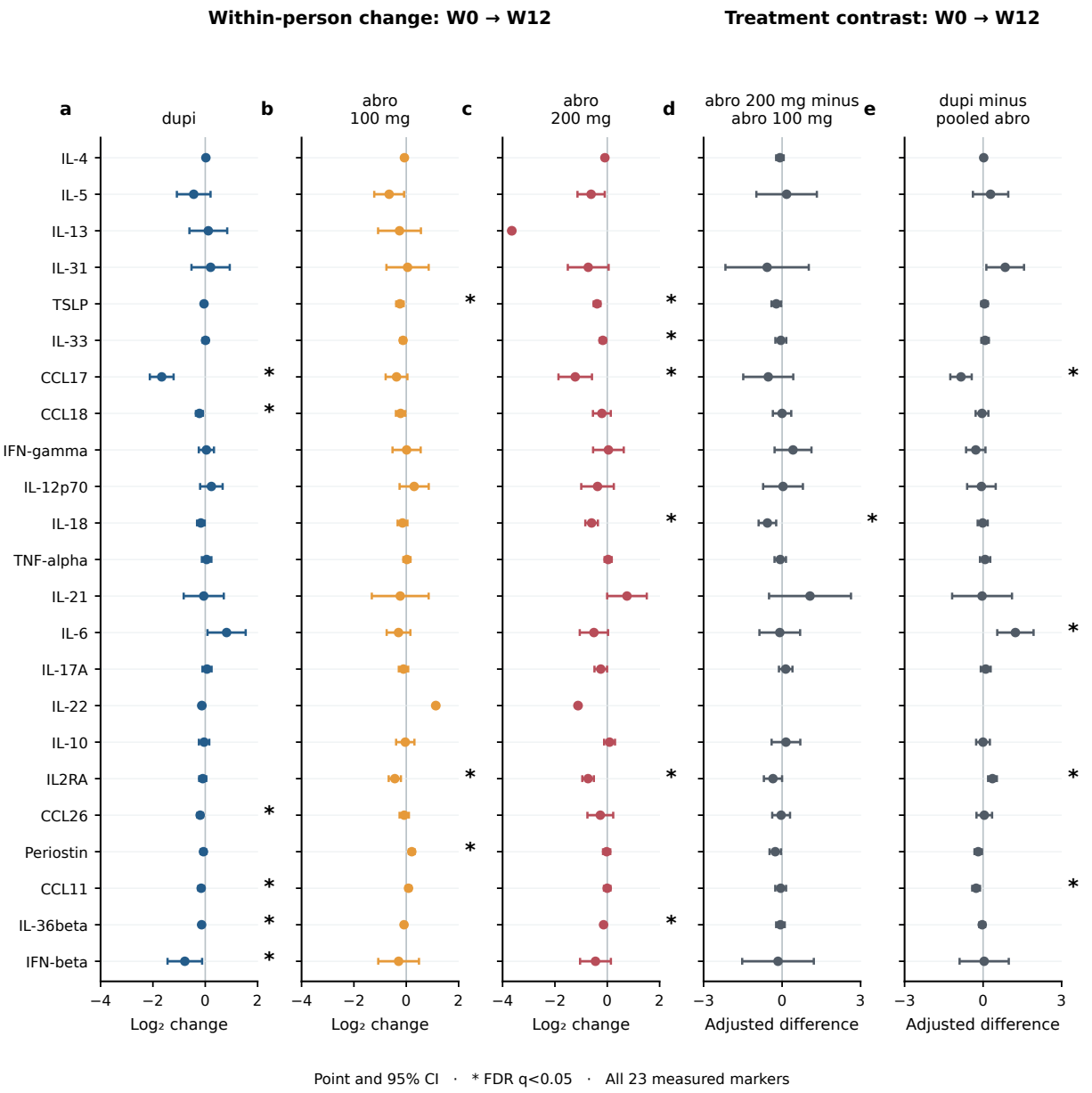


Figure 4: **Serum molecular response during the first 12 weeks.** a–c, Mean within-patient change in $\log_2$ concentration from W0 to W12 and 95% t confidence intervals for all 23 markers, by initial treatment. A change of −1 corresponds to half the baseline concentration. d–e, Direct adjusted differences in those changes for abro 200 mg minus abro 100 mg and dupi minus pooled abro, with HC3-robust 95% confidence intervals. Models adjust for starting $\log_2$ concentration, W0 EASI, age, sex, BMI and centre. pooled abro preserves the observed dose mixture. Negative contrasts mean a larger decline in the first group. Stars show q<0.05 under the original declared BH families; no tests or multiplicity families were changed for this display. Each marker uses its own available pairs. Estimates with fewer than five pairs have no inferential interval or star; missing estimates are left blank. Exact sample sizes, estimates, intervals and q values remain in source tables. Within-group and between-group panels have separately labelled x axes and cannot be interpreted as the same effect. The remaining pairwise treatment contrasts, later windows, fixed three-visit trajectories and assay-batch sensitivities are reported in the supplement. Treatment assignment was observational.

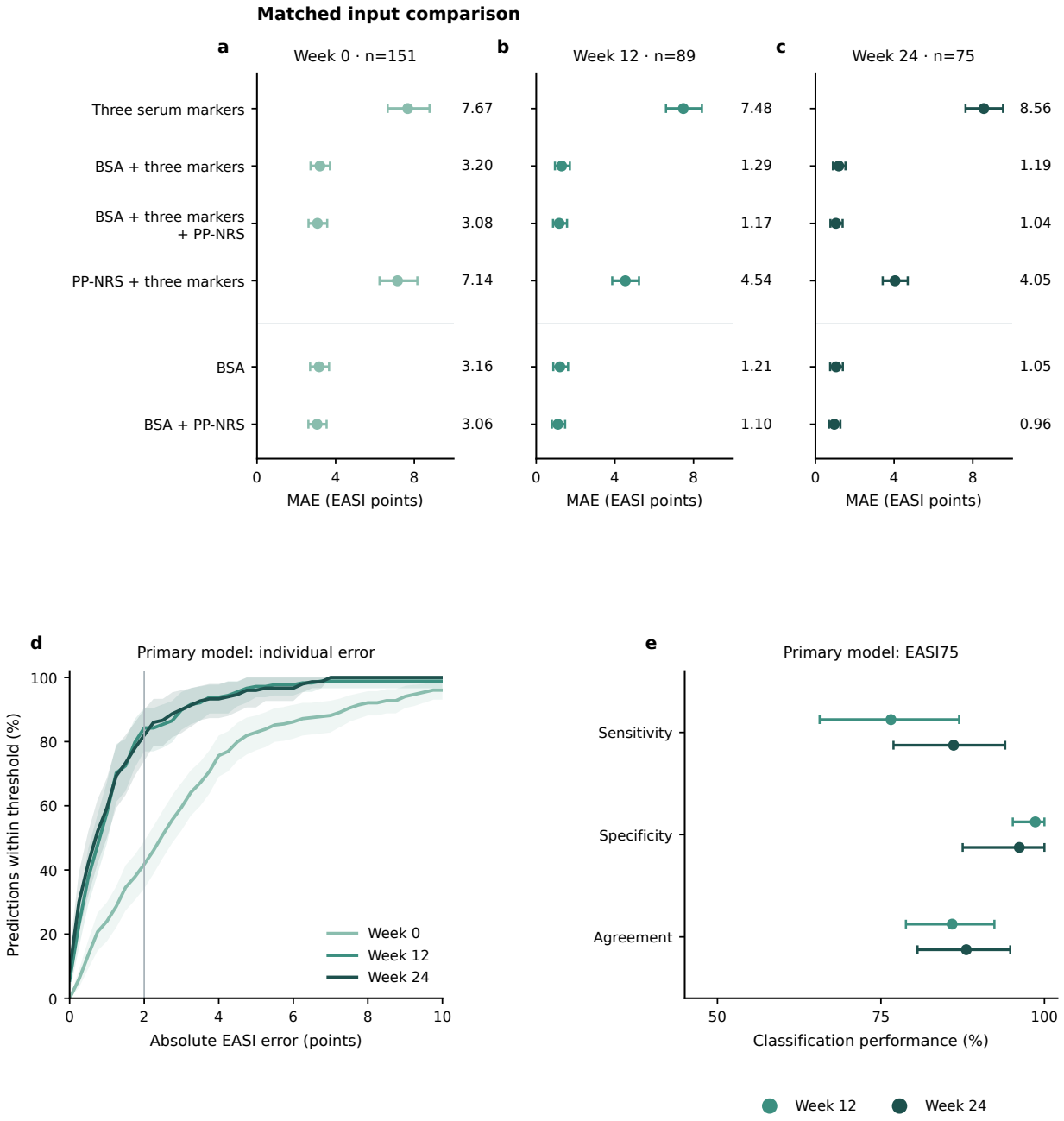


Figure 5: **Input contributions and performance of the common EASI estimator.** a–c, Post hoc input-set comparisons on identical patients with all three serum markers, BSA, PP-NRS and observed EASI. W0/W12/W24 contain 151/89/75 visits. Each model was refitted using the same patient outer folds; W0/W12 alone supplied training data. Preprocessing, structure and penalty selection used training folds. Serum markers are CCL17, CCL26 and IL-5; clinical inputs are from the current visit. Points and labels show mean absolute error (MAE), with 95% paired-patient bootstrap intervals. Clinical-only controls appear below the divider. Lower MAE after adding BSA to markers does not demonstrate molecular added accuracy; the relevant molecular comparisons retain the same clinical inputs. d–e, The unchanged primary BSA–serum procedure, evaluated on its original population. d, Cumulative absolute-error proportions among 152/89/75 patients, with bootstrap 95% intervals. All errors enter the calculation; only thresholds up to ten points are displayed. Two points is a visual reference, not an accepted interchangeability margin. e, EASI75 sensitivity, specificity and agreement, with bootstrap 95% intervals; W12/W24 responder denominators are 49/54, nonresponder denominators 36/13, and total paired denominators 85/67. Observed and estimated responses each use their own baseline and follow-up scores. All metrics are scored within each repetition before averaging. Bootstrap intervals condition on saved predictions. W24 was previously inspected and is same-cohort temporal application. Additional molecular-panel and adult-DLQI comparisons are in Supplementary Table S6 and its complete source tables. The new input comparisons do not replace the frozen primary equation.

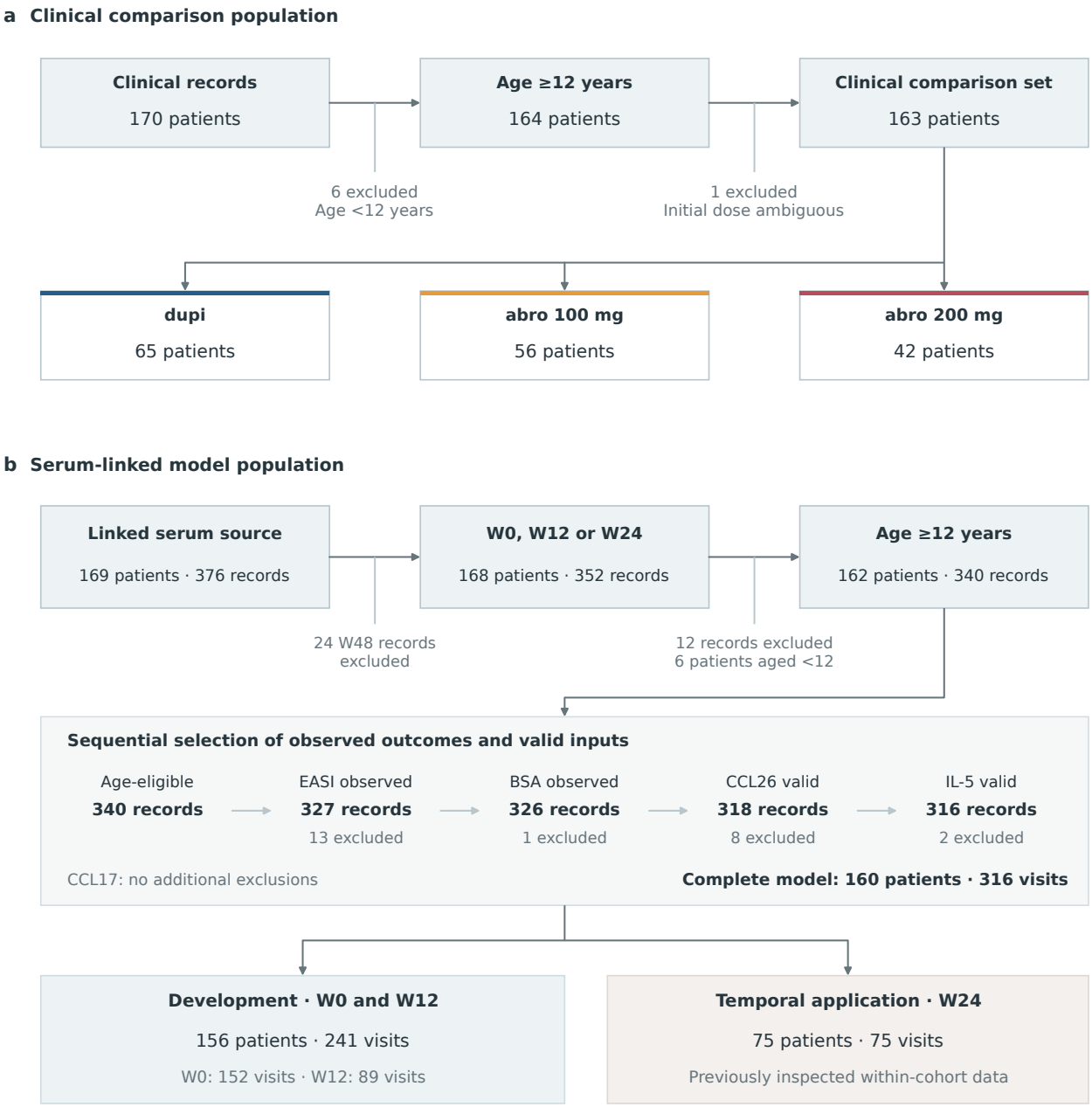


Figure S1: **Patient and record flow into the clinical and model analyses.** a, The curated clinical source contains 170 patients, not a screened or consented recruitment denominator. Six patients aged <12 years and one age-eligible patient with an ambiguous initial dose are excluded, leaving 163 patients classified by initial treatment: dupi, 65; abro 100 mg, 56; and abro 200 mg, 42. b, The linked serum source is a separate entry branch; model eligibility is not conditional on membership of the clinical comparison population. Removing 24 W48 records leaves 168 patients with 352 W0/W12/W24 records; age exclusions remove 12 records from six patients. Further ordered exclusions are mutually exclusive at the record level: 13 EASI, one BSA, zero CCL17, eight CCL26 and two IL-5 inputs are unavailable, leaving 316 complete model visits from 160 patients. A valid molecular input requires an available positive concentration for the log2 transformation. Loss of a record does not necessarily remove a patient. Development uses 241 W0/W12 visits from 156 patients, with patient-level validation splits; W24 includes 75 visits from 75 patients and is previously inspected within-cohort temporal application, not external validation. Development and W24 share 71 patients, with 85 development-only and four W24-only patients. Fifty-one patients have all three complete visits. The clinical and complete-input model populations share 159 patients, with four clinical-only and one model-only patient; the treatment-free model can retain an ambiguous initial dose. Scheduled empty visit slots are not counted as observed visits, and no filled-in clinical outcome is used as validation truth.

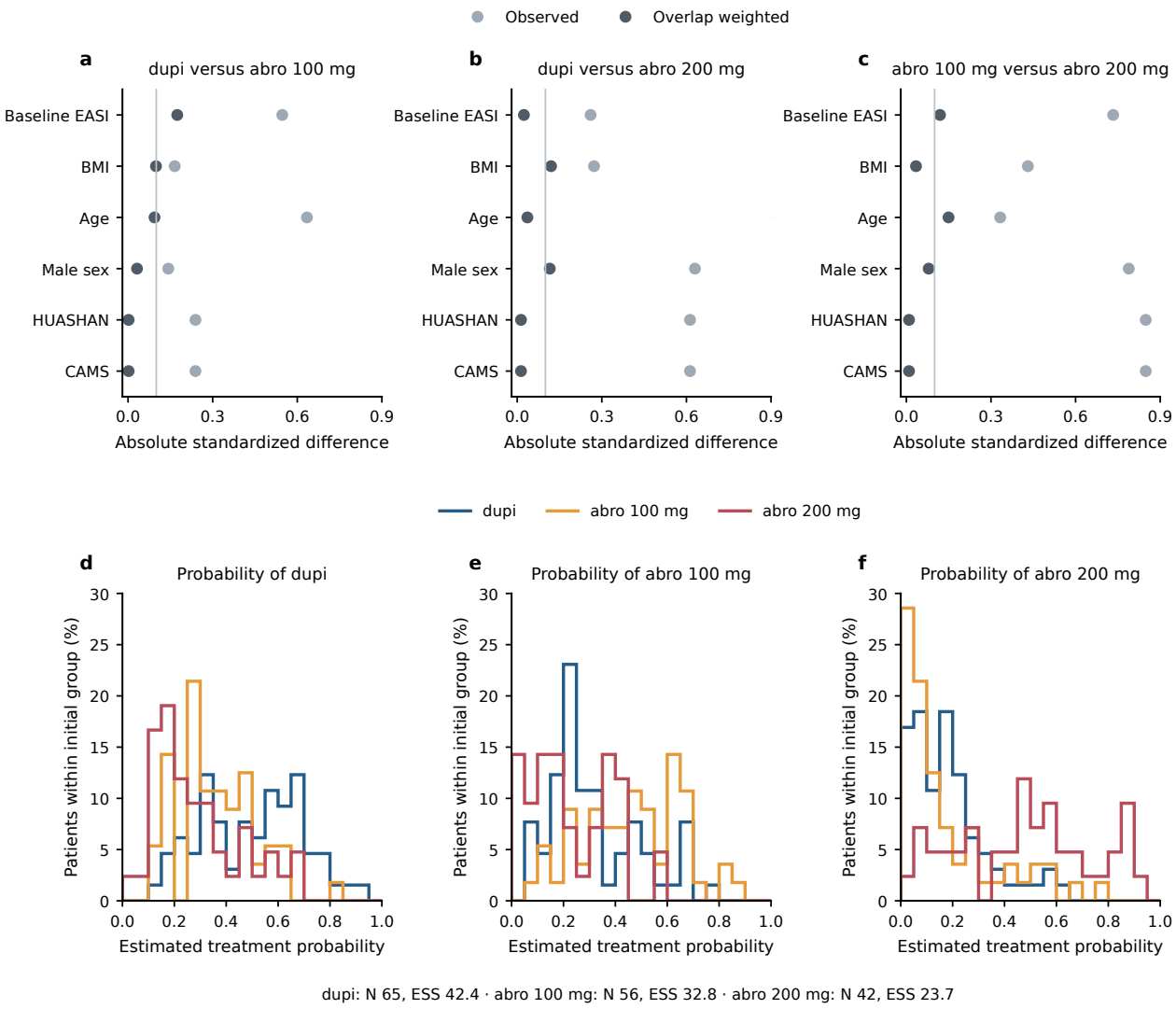


Figure S2: **Baseline balance, treatment overlap and effective sample size.** a–c, Absolute standardized mean differences for every covariate in the initial-treatment propensity model, before and after generalized overlap weighting. Differences use the unweighted pairwise pooled standard deviation in both displays. The light 0.10 reference line is a descriptive balance guide, not a statistical significance threshold. Residual imbalance remains. d–f, Baseline multinomial treatment probabilities, stratified by actual initial treatment; each distribution uses all patients in that initial group and sums to 100%. These show common support rather than longitudinal drug exposure. The footer reports original N and effective sample size, $(\text{sum w})^2/\text{sum}(\text{w}^2)$. All centre, sex, age, BMI and EASI variables are shown. Overlap weighting targets comparable baseline patients; it cannot remove unmeasured confounding or correct outcome-driven dose changes. Endpoint- and marker-specific availability/ESS diagnostics are retained in the source tables.

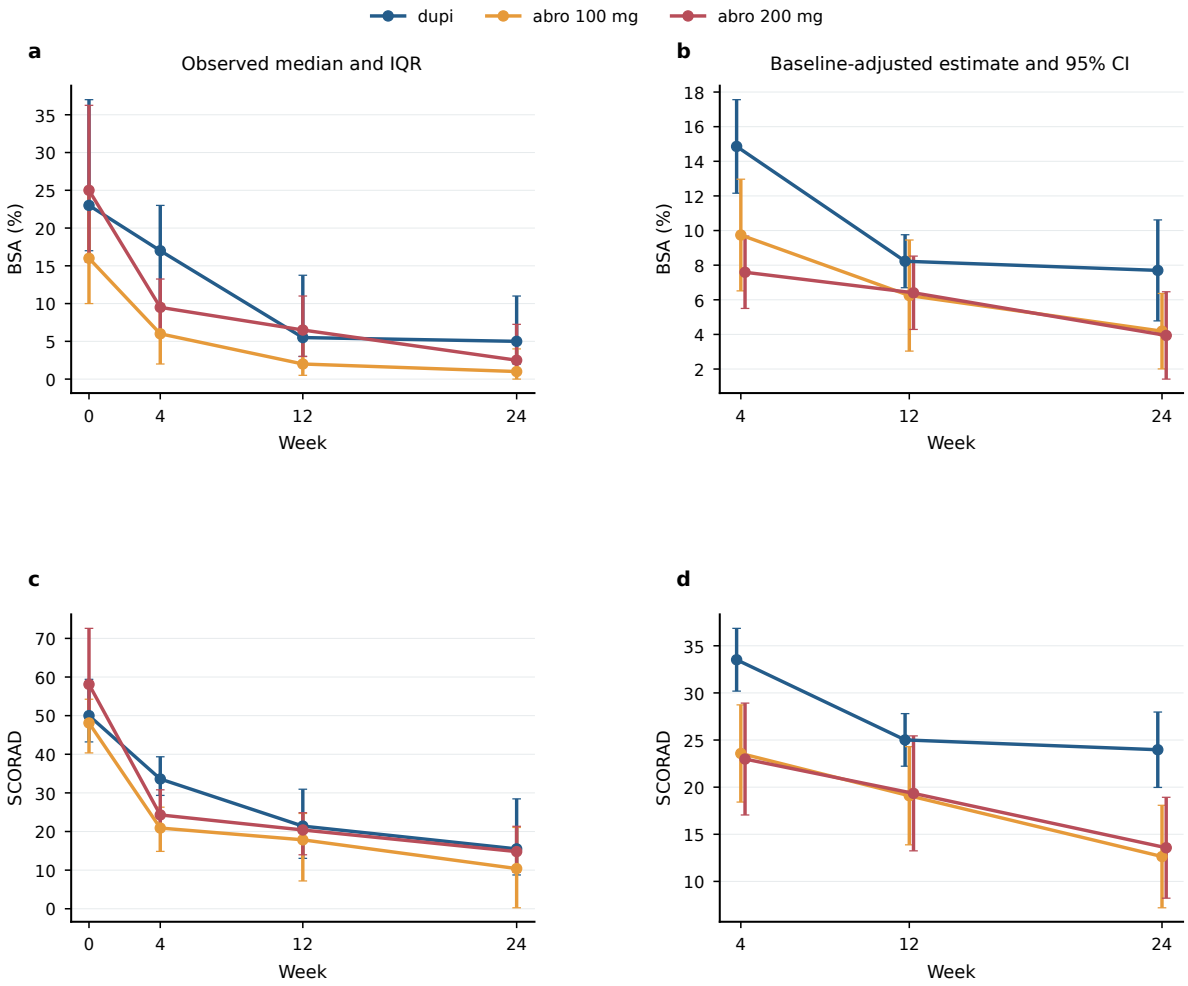


Figure S3: **Observed and adjusted BSA (%), SCORAD.** Left, observed medians with interquartile ranges by initial treatment; all available values are used at each specified visit. Right, generalized-overlap-weighted marginal estimates with patient-cluster robust 95% confidence intervals, adjusted for baseline outcome, age, sex, BMI and centre. Patient sets can differ between visits. Exact outcome-specific counts and numerical comparisons are in the source tables; they are not replaced by a common denominator. Initial treatment groups describe observational management strategies, including subsequent regimen changes, rather than randomized fixed-dose effects.

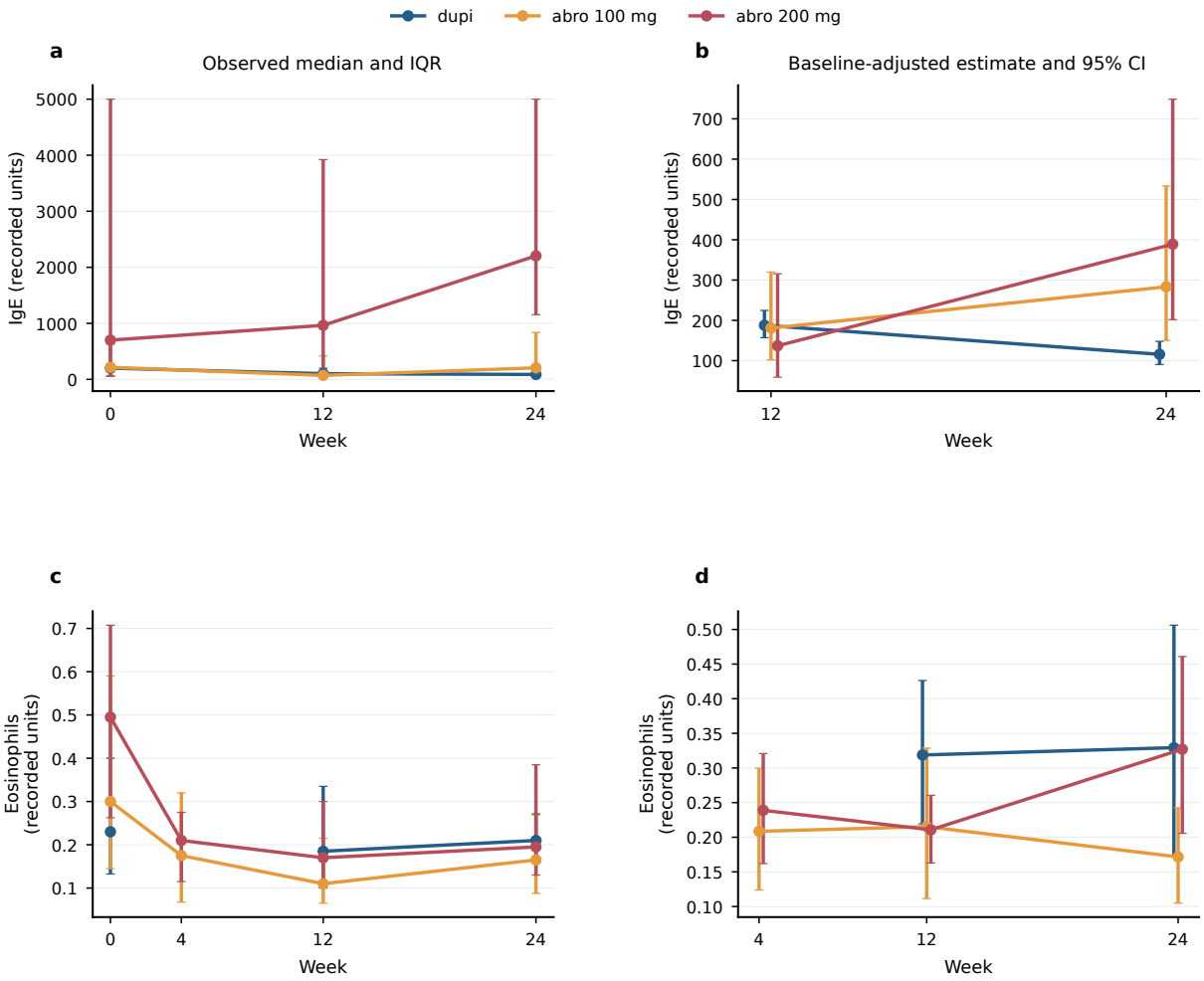


Figure S4: **Observed and adjusted IgE (recorded units), Eosinophils (recorded units).** Left, observed medians with interquartile ranges by initial treatment; all available values are used at each specified visit. Right, generalized-overlap-weighted marginal estimates with patient-cluster robust 95% confidence intervals, adjusted for baseline outcome, age, sex, BMI and centre. Patient sets can differ between visits. Exact outcome-specific counts and numerical comparisons are in the source tables; they are not replaced by a common denominator. IgE and eosinophils use log(1+x) models and back-transformation; adjusted estimates are not arithmetic means. IgE is shown only at W0/W12/W24. No dupi W4 eosinophil measurement is available; both the observed and adjusted line break at that visit. No multiple-imputation value is displayed as an observation. The separate same-patient IgE analysis tests the effect of changing sample composition. Initial treatment groups describe observational management strategies, including subsequent regimen changes, rather than randomized fixed-dose effects.

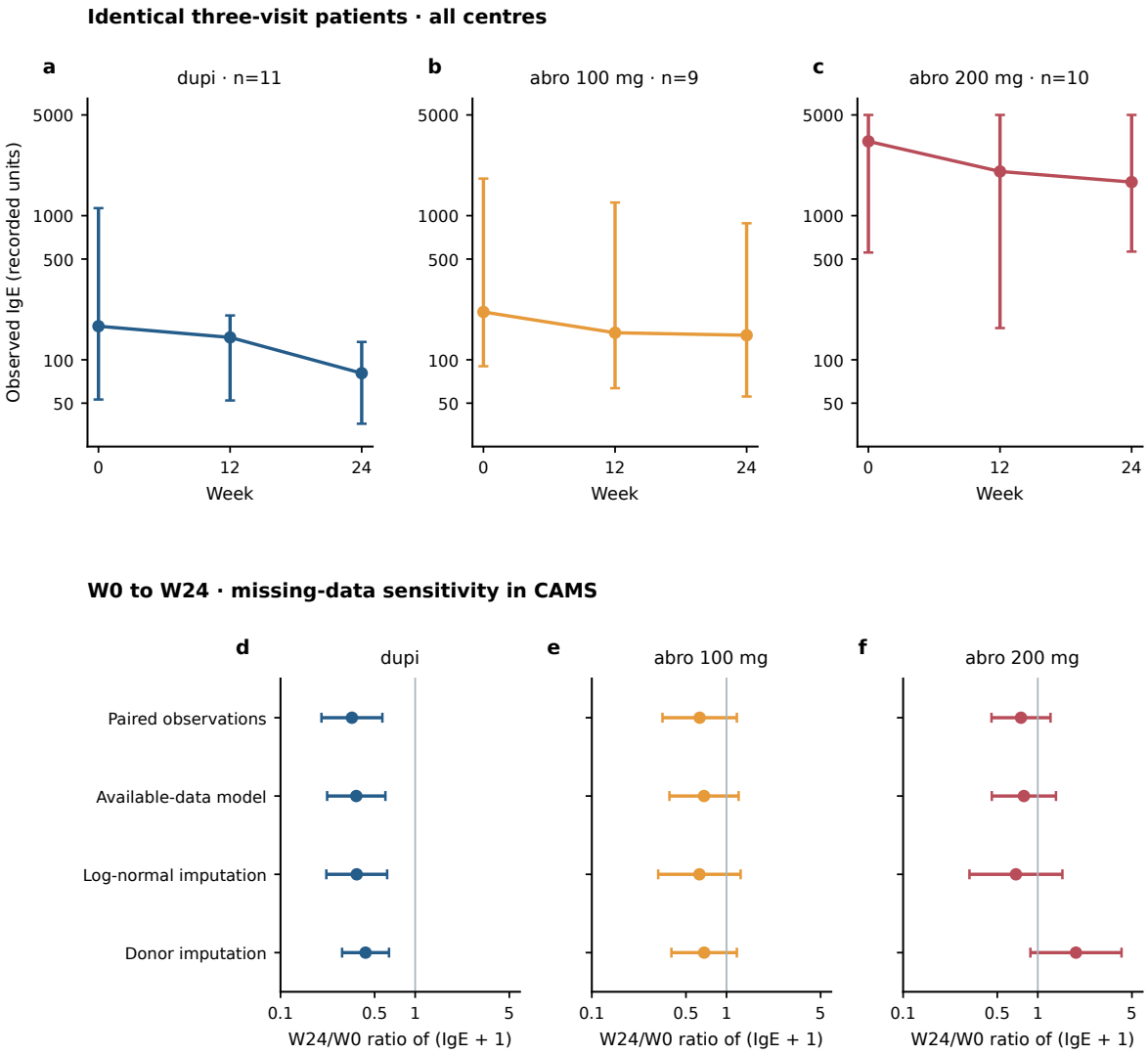


Figure S5: **IgE: fixed-patient observations and missing-data sensitivity.** a–c, Observed medians and patient-bootstrap 95% confidence intervals in the same complete three-visit patients at every visit: 11 dupi, 9 abro 100 mg and 10 abro 200 mg patients across both centres. The vertical scale is logarithmic, labelled in recorded units. d–f, W0–W24 geometric mean ratios of IgE+1 from four analyses within CAMS. The observed paired analysis uses all available W0/W24 pairs (14/10/12 by treatment); it does not require W12. The available-data random-intercept model uses 229 measured visits from 125 patients across the three treatments. Multiple-imputation analyses target all 130 CAMS patients (51/38/41), using patient-wide chained equations with either posterior log-normal draws or within-treatment predictive mean matching (donor imputation); 400 retained datasets are pooled with Rubin variance and finite-sample degrees of freedom. Both assume missingness can be explained by the included observed information. The panels target the same direction of change but rely on different populations or missingness assumptions. Ratios below one indicate lower W24 values; the line at one denotes no change. Bars are 95% confidence intervals. No observed value was replaced. Sparse donor support makes the abro 200 mg estimate sensitive to the imputation method; neither imputation confirms its direction of change. CAMS is used because Huashan has no follow-up IgE for dupi. Full-cohort, other-window and missing-not-at-random analyses are retained in Supplementary Table S8 and source data. No W4 IgE was observed, so no W4 point is imputed or plotted. The recorded value 5000 was not assumed to be a verified censoring threshold. These observational analyses do not identify causal drug effects.

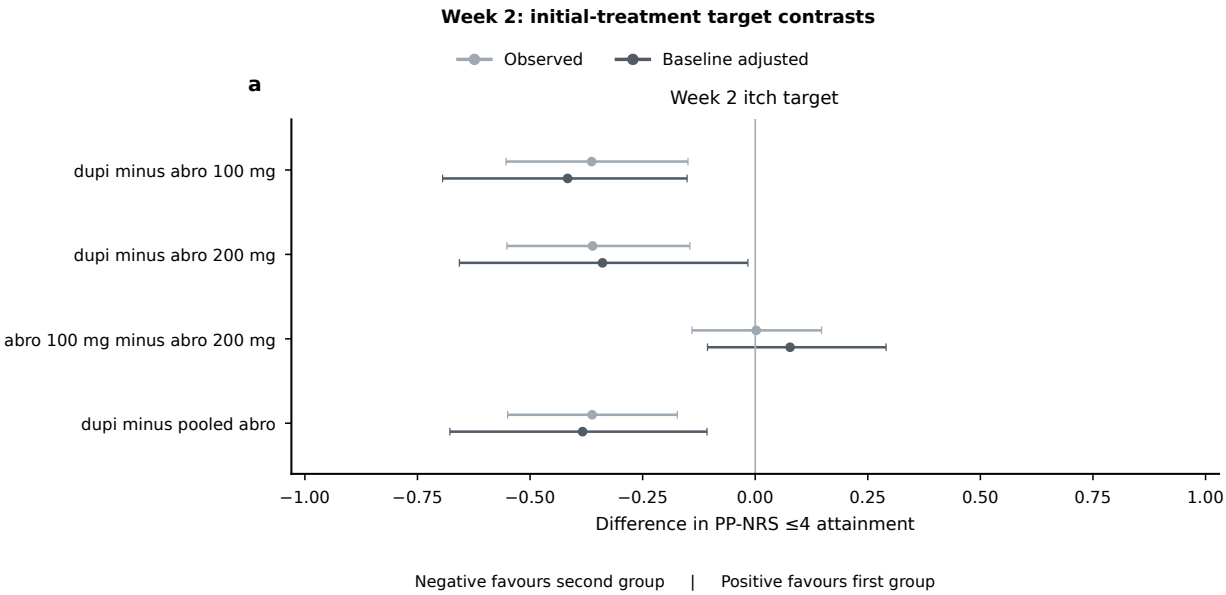


Figure S6: **Week 2: all initial-treatment target contrasts.** Observed and baseline-adjusted risk differences with nominal 95% confidence intervals. Positive values favour the first named treatment for target attainment. Adjustment uses generalized overlap weights from baseline EASI, age, sex, BMI and centre; it does not estimate a randomized current-dose effect. Treatment labels refer to the initial prescribed regimen. pooled abro is supplementary. Missing estimates are left absent; all exact denominators, Fisher/weighted tests, Holm-adjusted P values and exploratory FDR values are in the source tables.

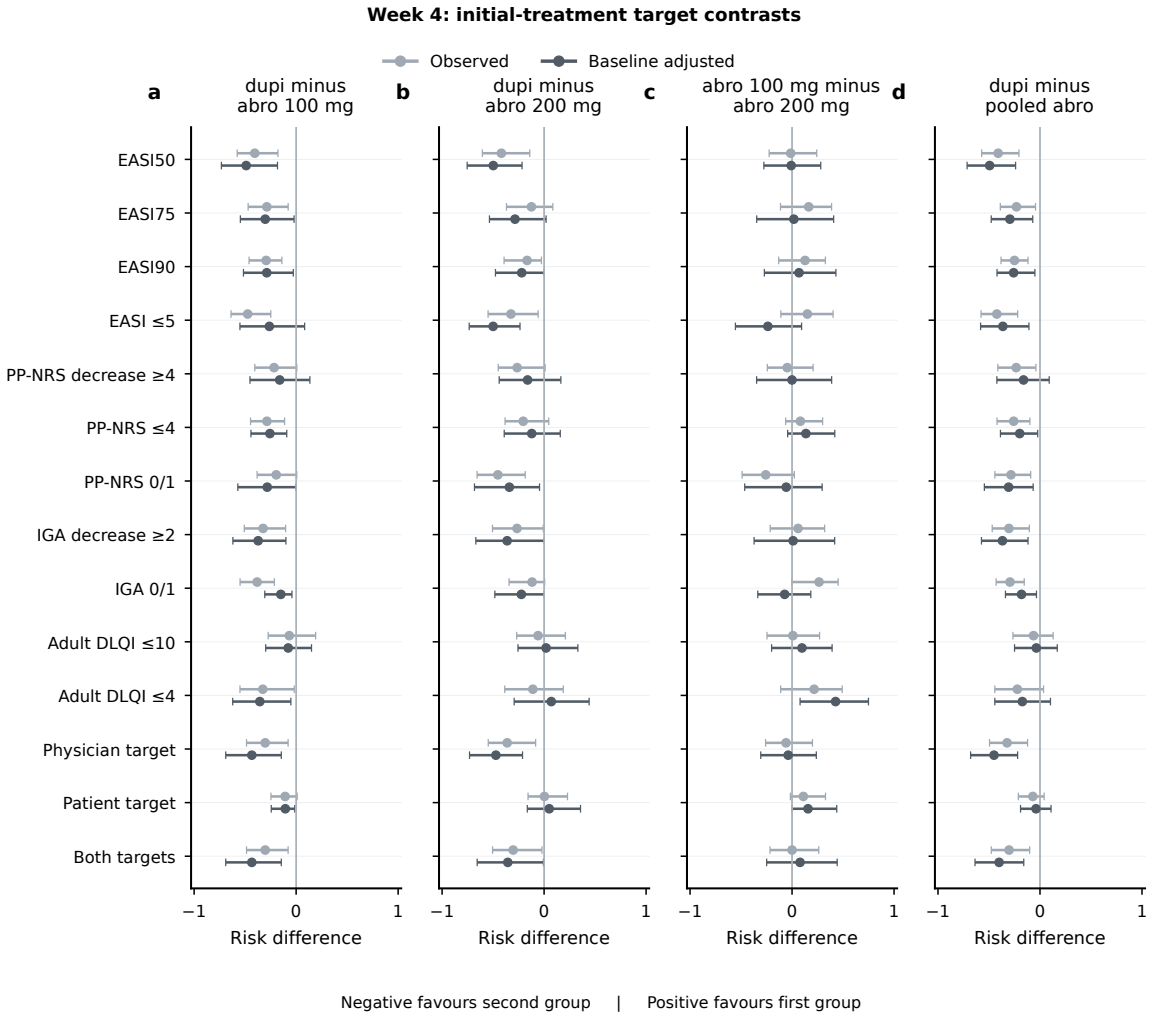


Figure S7: **Week 4: all initial-treatment target contrasts.** Observed and baseline-adjusted risk differences with nominal 95% confidence intervals. Positive values favour the first named treatment for target attainment. Adjustment uses generalized overlap weights from baseline EASI, age, sex, BMI and centre; it does not estimate a randomized current-dose effect. Treatment labels refer to the initial prescribed regimen. pooled abro is supplementary. Missing estimates are left absent; all exact denominators, Fisher/weighted tests, Holm-adjusted P values and exploratory FDR values are in the source tables.

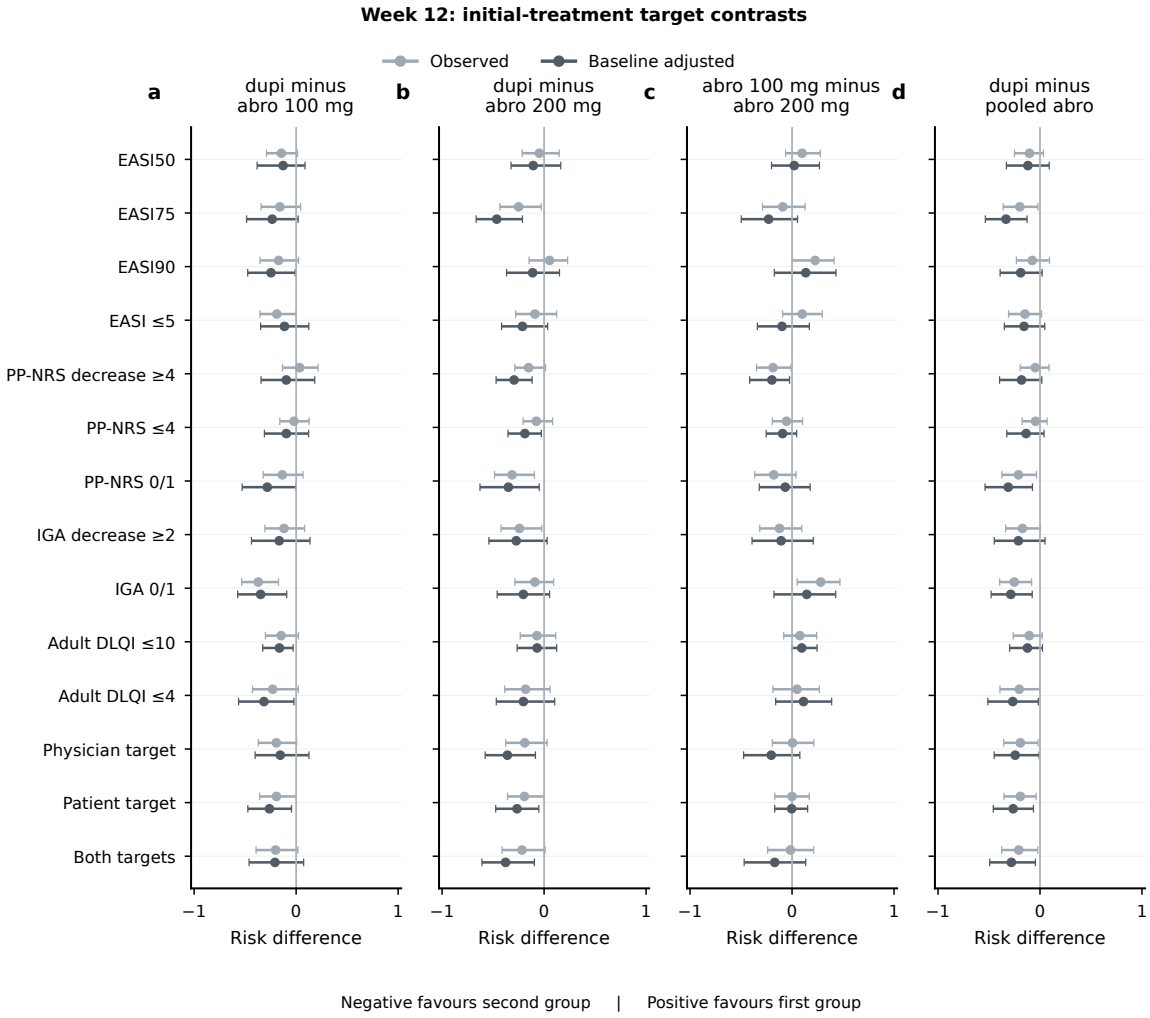


Figure S8: **Week 12: all initial-treatment target contrasts.** Observed and baseline-adjusted risk differences with nominal 95% confidence intervals. Positive values favour the first named treatment for target attainment. Adjustment uses generalized overlap weights from baseline EASI, age, sex, BMI and centre; it does not estimate a randomized current-dose effect. Treatment labels refer to the initial prescribed regimen. pooled abro is supplementary. Missing estimates are left absent; all exact denominators, Fisher/weighted tests, Holm-adjusted P values and exploratory FDR values are in the source tables.

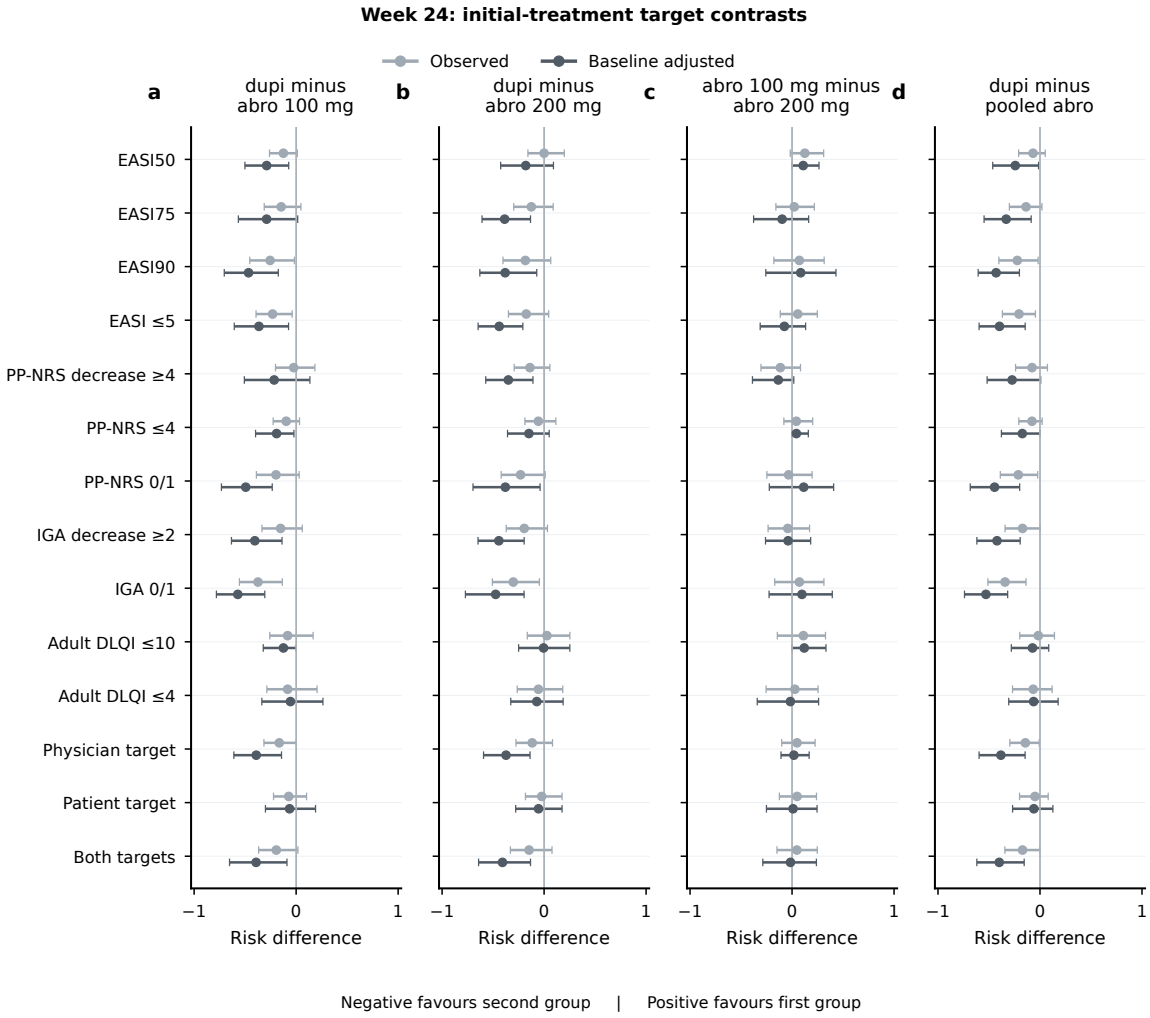


Figure S9: **Week 24: all initial-treatment target contrasts.** Observed and baseline-adjusted risk differences with nominal 95% confidence intervals. Positive values favour the first named treatment for target attainment. Adjustment uses generalized overlap weights from baseline EASI, age, sex, BMI and centre; it does not estimate a randomized current-dose effect. Treatment labels refer to the initial prescribed regimen. pooled abro is supplementary. Missing estimates are left absent; all exact denominators, Fisher/weighted tests, Holm-adjusted P values and exploratory FDR values are in the source tables.

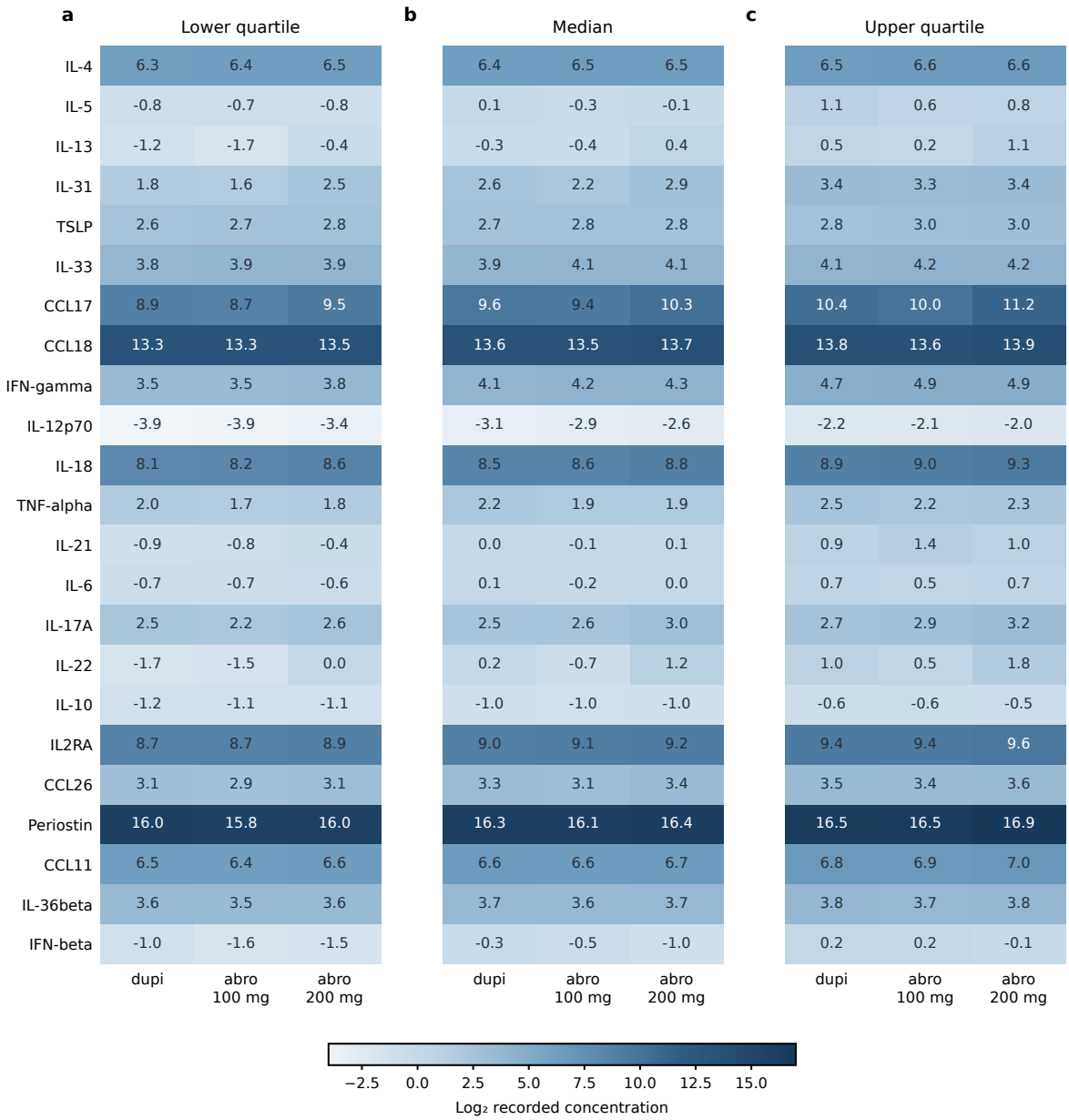


Figure S10: **Baseline distributions of all 23 serum markers.** Cells display the lower quartile, median and upper quartile of baseline $\log_2$ recorded concentration for each initial treatment. All three panels use the same numeric colour scale without row normalization. Compare treatments within a marker: concentration differences between distinct analytes do not indicate differences in biological activity. dupi denotes dupi; abro 100 mg/abro 200 mg initial abro 100/200 mg. Every measured marker is retained. The source table includes exact counts, censoring-related availability, ranges and distribution summaries; quartiles summarize available positive numeric observations and do not replace censored values.

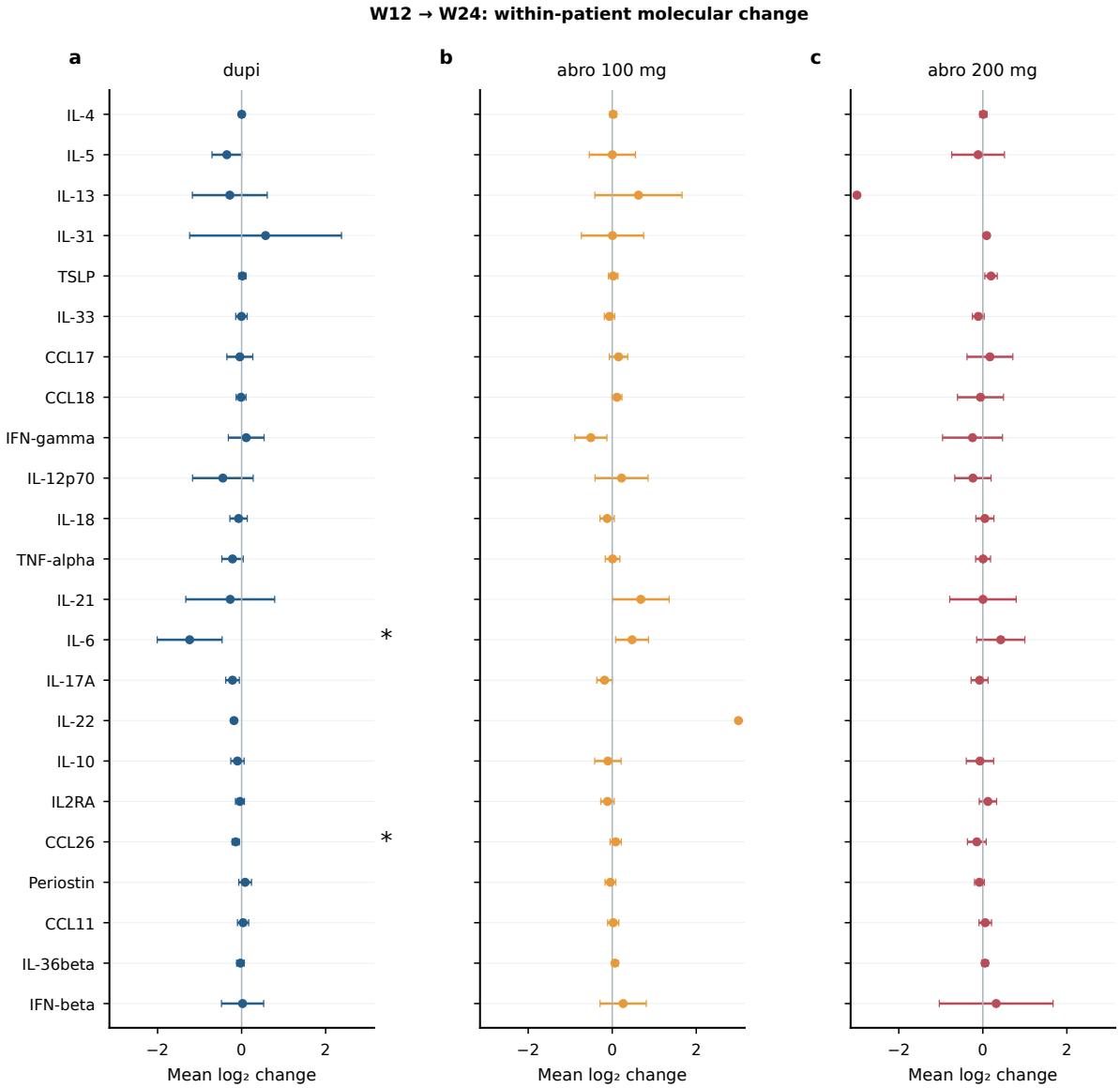


Figure S11: **Molecular changes from W12 to W24.** Mean paired $\log_2$ concentration changes with 95% t confidence intervals. Negative values indicate lower later concentration. Each marker/treatment uses all available pairs for this interval; exact counts are in the source table. All 23 measured markers are retained. Estimates based on fewer than five pairs are descriptive and have no inferential interval; unavailable estimates are blank. Stars denote BH q<0.05 across the full 207 marker-by-treatment-by-window family, not a separate family for the displayed interval.

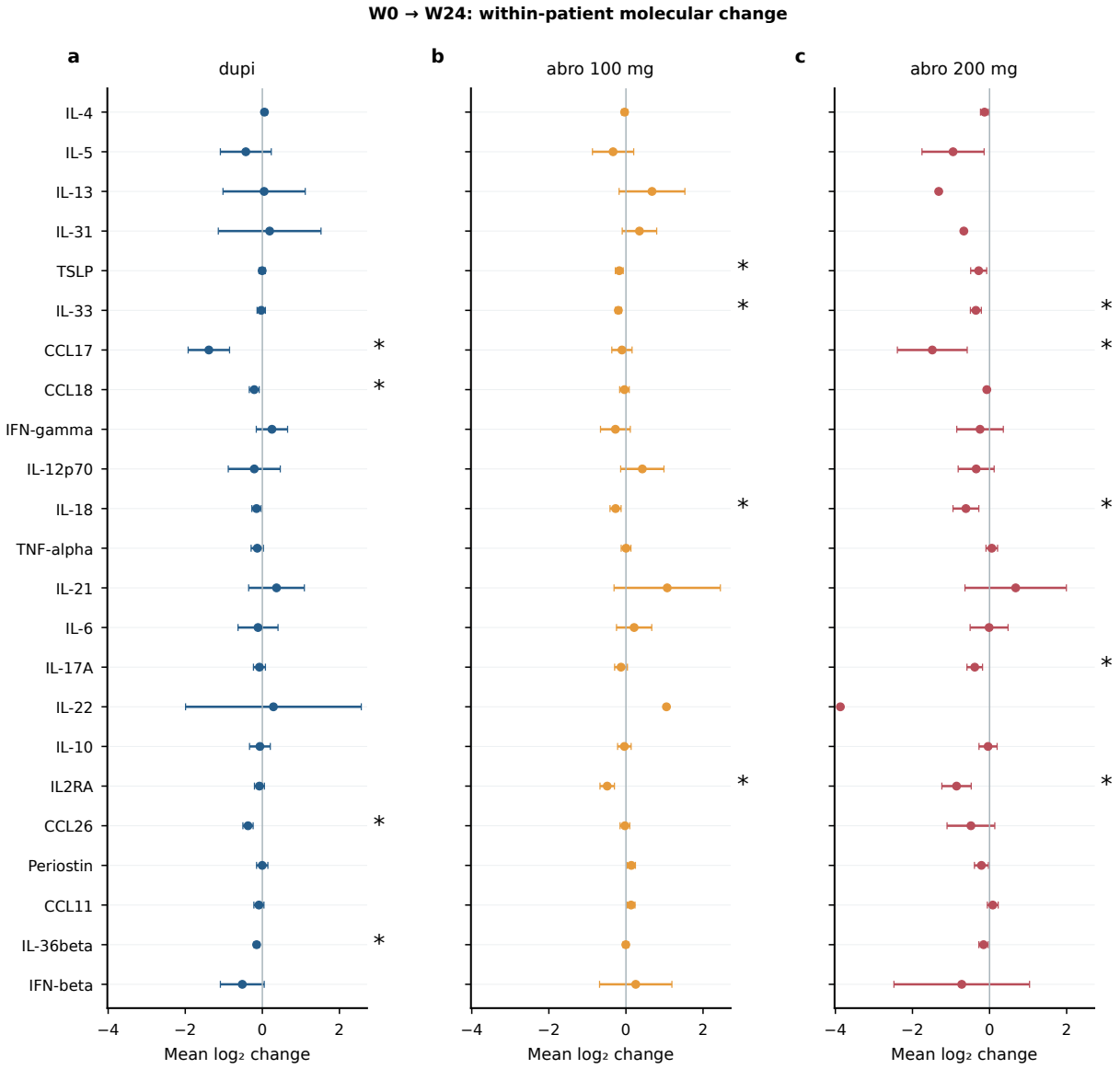


Figure S12: **Molecular changes from W0 to W24.** Mean paired $\log_2$ concentration changes with 95% t confidence intervals. Negative values indicate lower later concentration. Each marker/treatment uses all available pairs for this interval; exact counts are in the source table. All 23 measured markers are retained. Estimates based on fewer than five pairs are descriptive and have no inferential interval; unavailable estimates are blank. Stars denote BH q<0.05 across the full 207 marker-by-treatment-by-window family, not a separate family for the displayed interval.

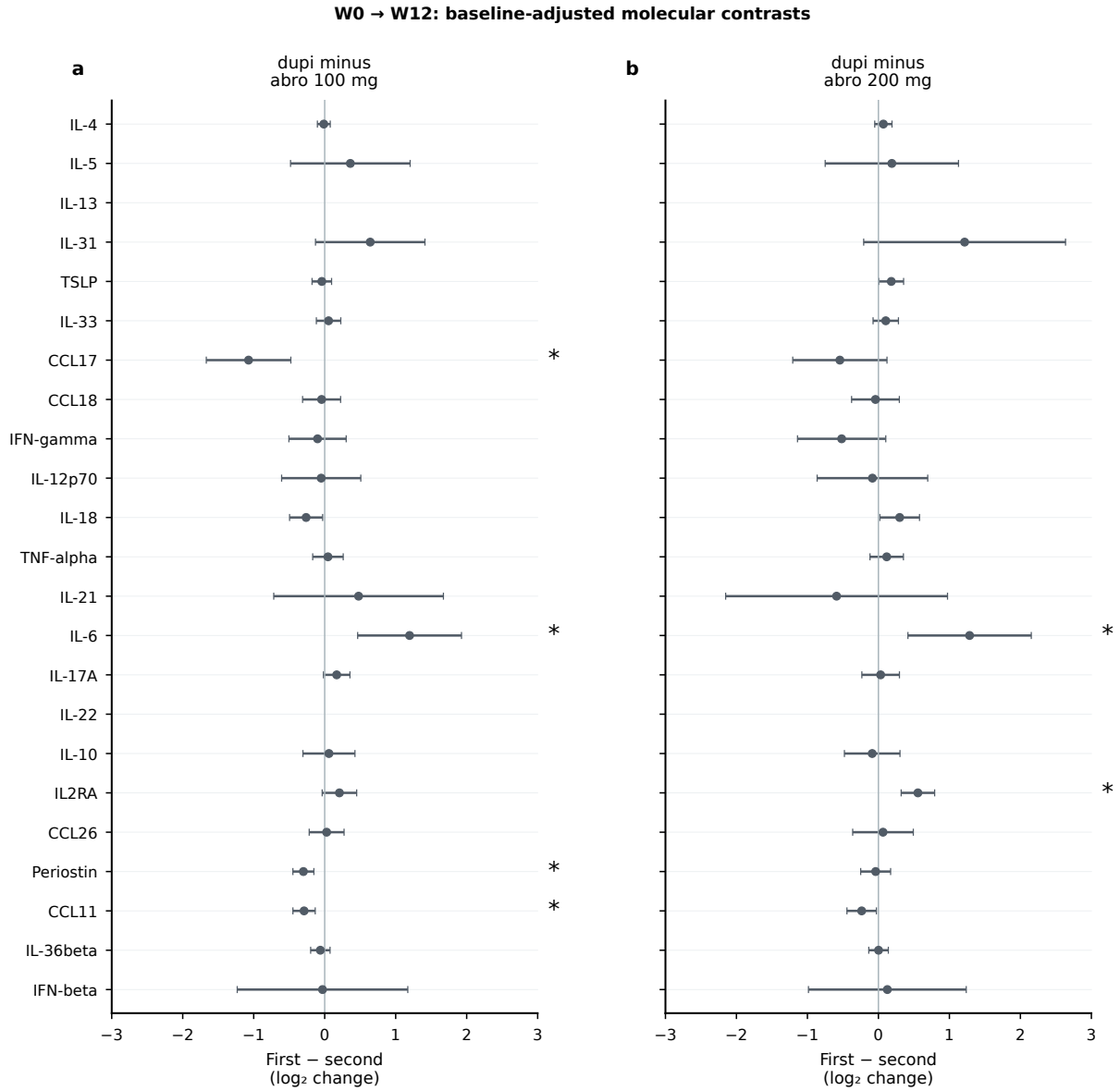


Figure S13: **Adjusted molecular comparisons from W0 to W12.** Differences in within-patient $\log_2$ changes with HC3 95% confidence intervals. Negative values mean a smaller, more negative change in the first named treatment. Models adjust starting $\log_2$ concentration, baseline total EASI, age, sex, BMI and centre. dupi, dupi; abro 100 mg/abro 200 mg, initial abro 100/200 mg. pooled abro retains the observed dose mixture. Stars denote the original BH q<0.05 family; no multiplicity family is recalculated after display selection. Blank positions have no estimable contrast. Complete counts, diagnostics, centre/batch sensitivities and tests remain in the source table. The W0–W12 dupi-minus-pooled-A and abro 200 mg-minus-abro 100 mg comparisons are in the main molecular figure. These are adjusted observational associations, not randomized current-dose effects.

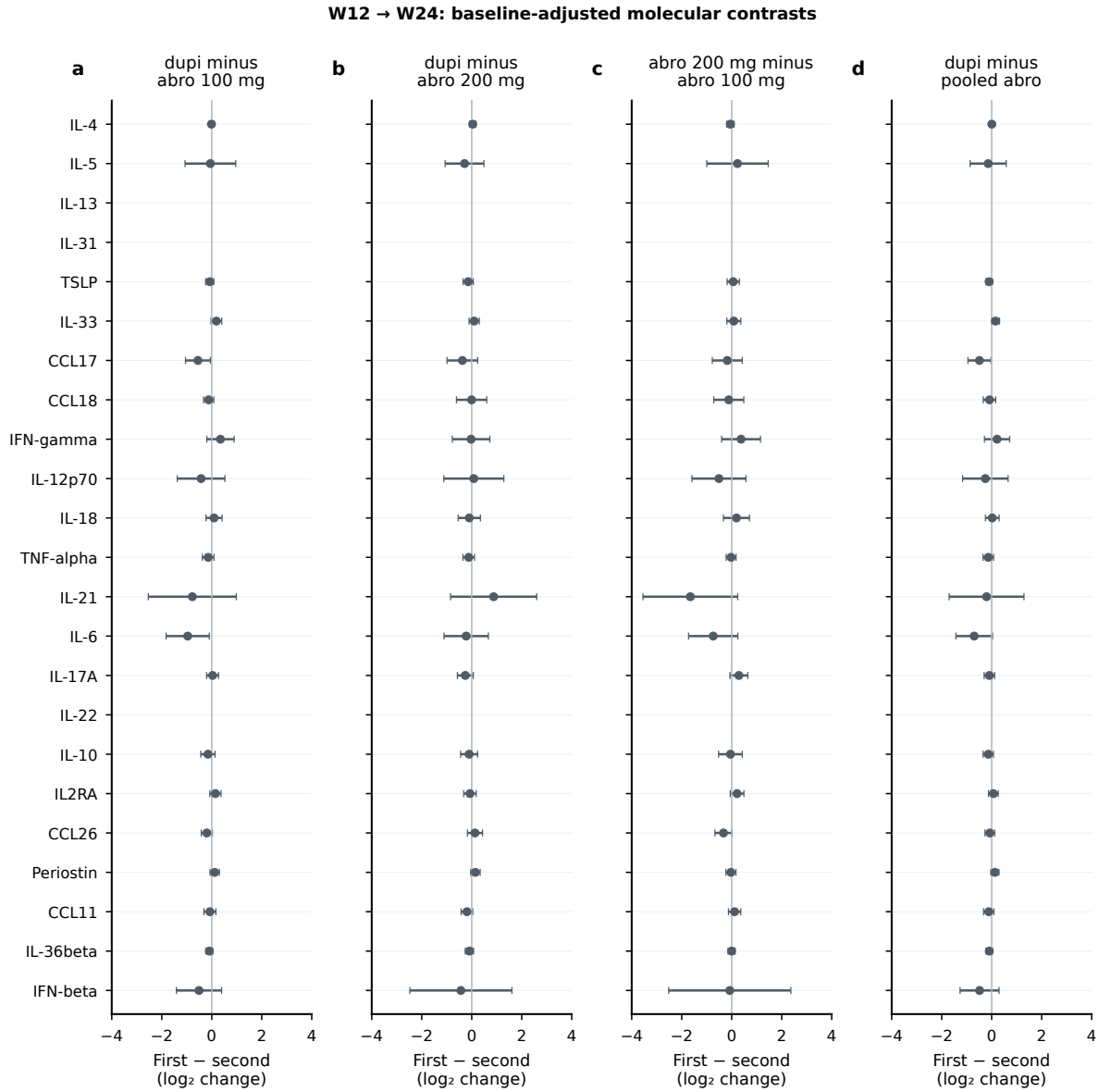


Figure S14: **Adjusted molecular comparisons from W12 to W24.** Differences in within-patient $\log_2$ changes with HC3 95% confidence intervals. Negative values mean a smaller, more negative change in the first named treatment. Models adjust starting $\log_2$ concentration, baseline total EASI, age, sex, BMI and centre. dupi, dupi; abro 100 mg/abro 200 mg, initial abro 100/200 mg. pooled abro retains the observed dose mixture. Stars denote the original BH q<0.05 family; no multiplicity family is recalculated after display selection. Blank positions have no estimable contrast. Complete counts, diagnostics, centre/batch sensitivities and tests remain in the source table. These are adjusted observational associations, not randomized current-dose effects.

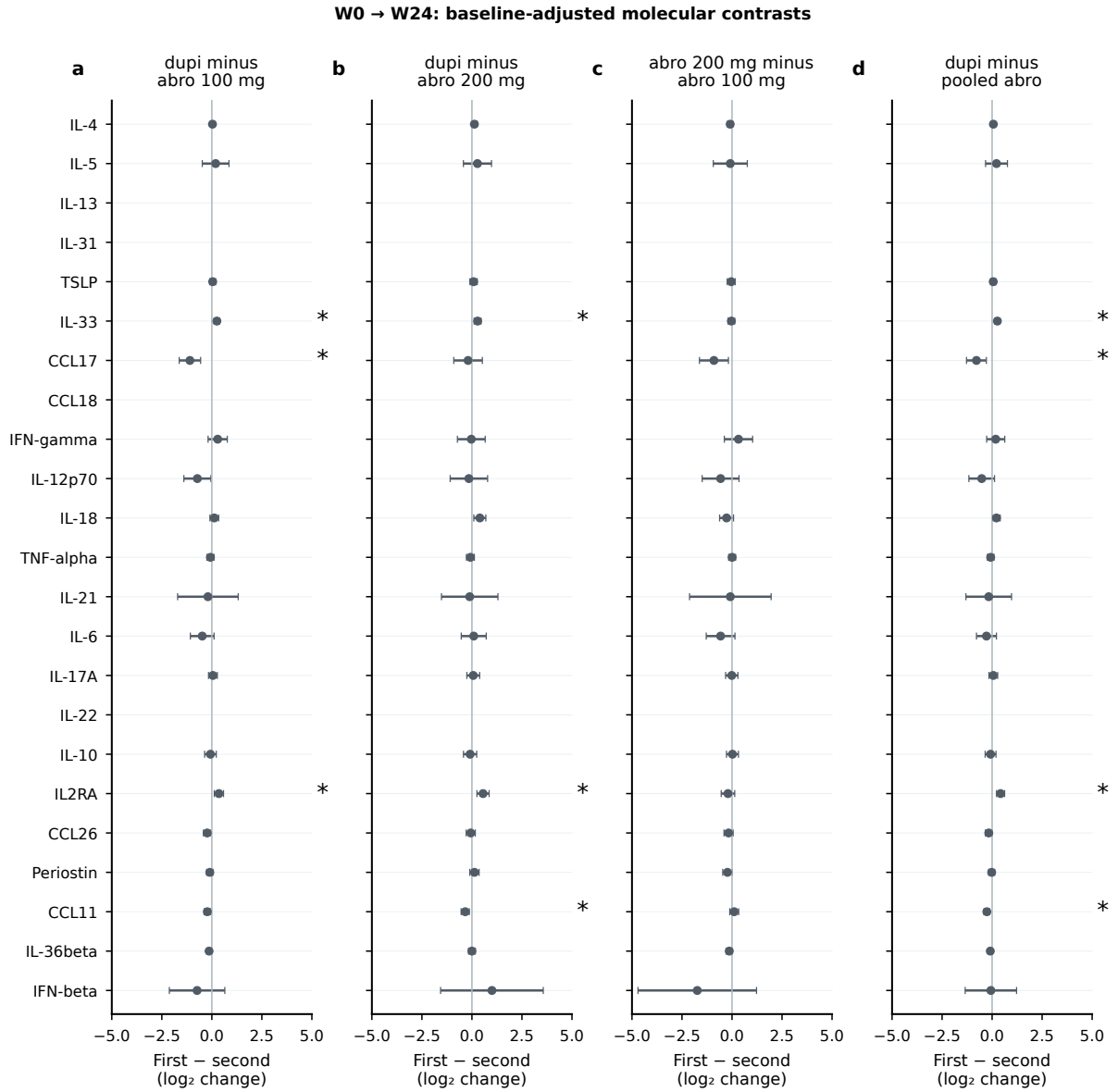


Figure S15: **Adjusted molecular comparisons from W0 to W24.** Differences in within-patient $\log_2$ changes with HC3 95% confidence intervals. Negative values mean a smaller, more negative change in the first named treatment. Models adjust starting $\log_2$ concentration, baseline total EASI, age, sex, BMI and centre. dupi, dupi; abro 100 mg/abro 200 mg, initial abro 100/200 mg. pooled abro retains the observed dose mixture. Stars denote the original BH q<0.05 family; no multiplicity family is recalculated after display selection. Blank positions have no estimable contrast. Complete counts, diagnostics, centre/batch sensitivities and tests remain in the source table. These are adjusted observational associations, not randomized current-dose effects.

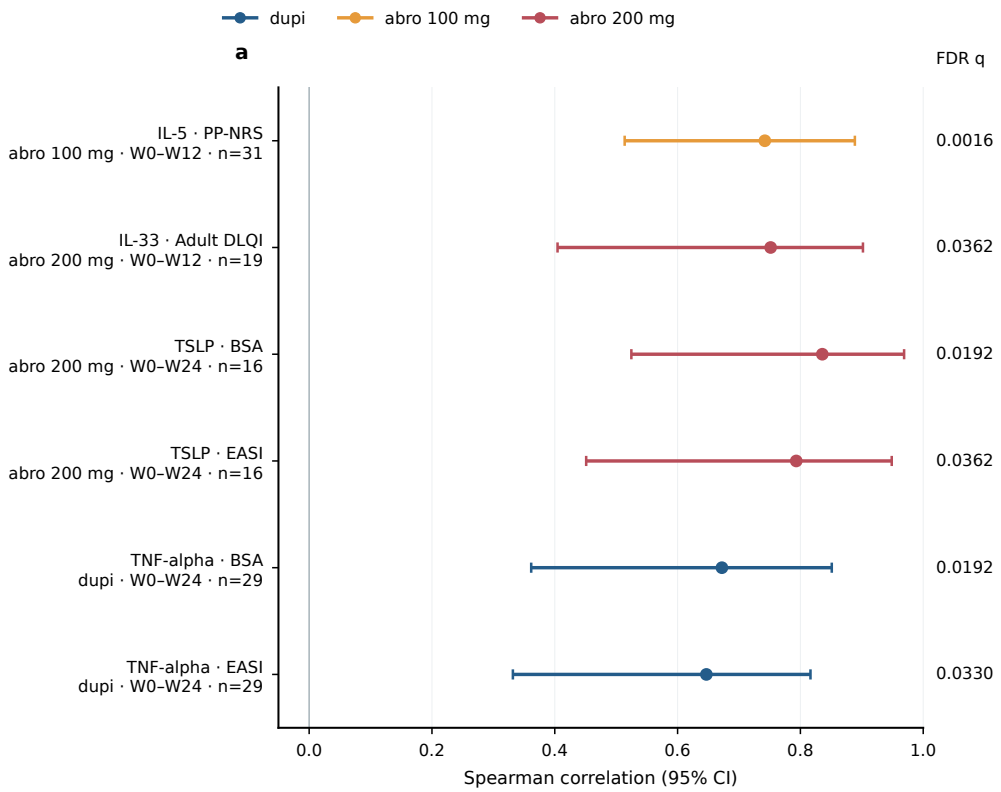


Figure S16: **Molecular and clinical changes associated after multiplicity correction.** Every association meeting BH q<0.05 in the full declared change-association family is displayed; selection spans 23 markers, five clinical outcomes, three initial treatments and three intervals. No W12–W24 association met this threshold. Points are Spearman correlations between within-patient clinical change and $\log_2$ molecular change, both later minus earlier, with patient-bootstrap 95% confidence intervals. Positive correlations indicate changes in the same direction. Labels give paired-patient counts and the interval; colours indicate initial treatment only. Adult DLQI requires age ≥18. All tested coefficients, non-significant results and unestimable pairs remain in the complete source table. This significance-selected display is exploratory and is not independent validation, numerical agreement or a causal molecular effect.

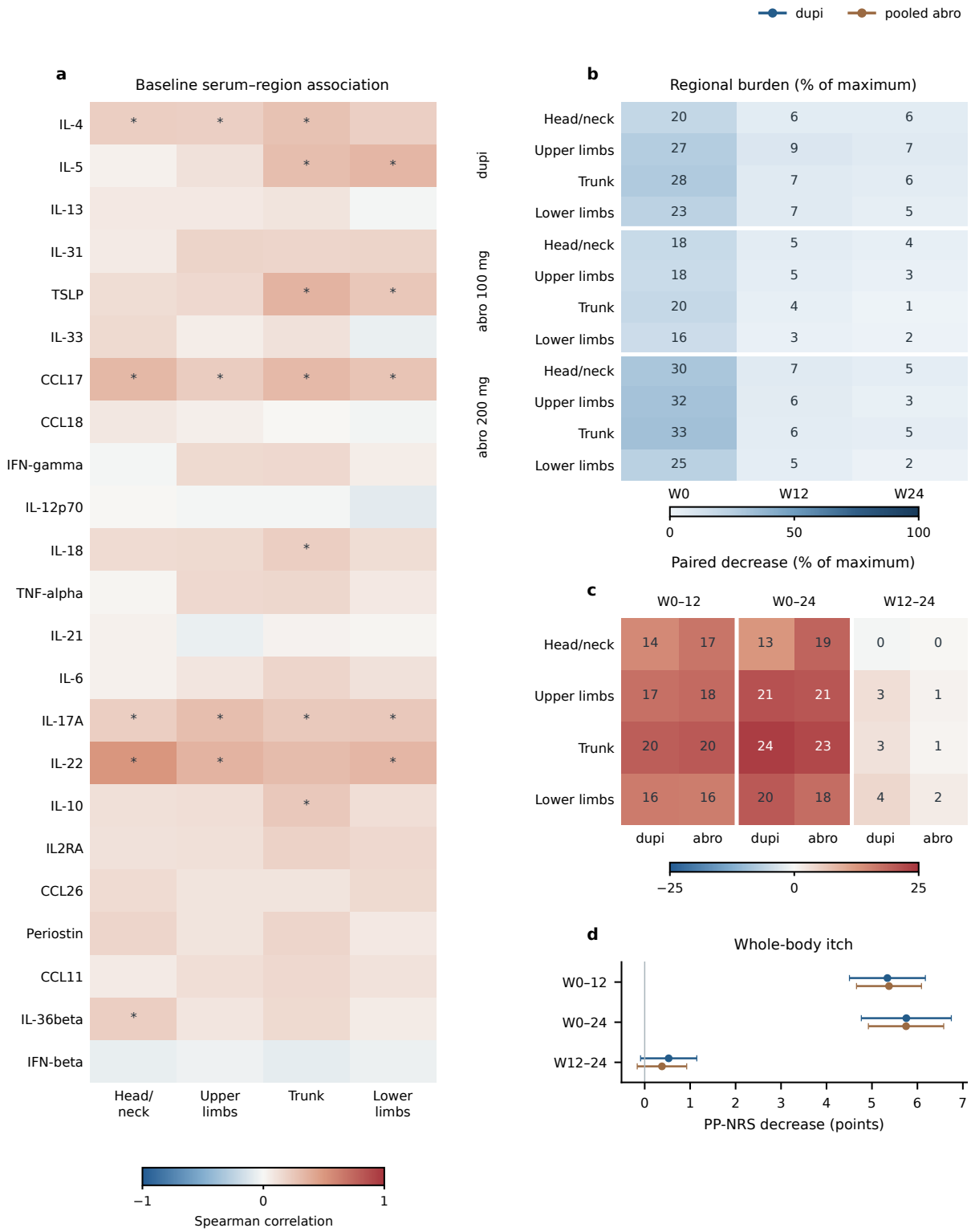


Figure S17: **Regional clinical burden and molecular associations.** a, Baseline Spearman associations between all 23 markers and the four regional EASI contributions; stars mark BH q<0.05, with complete coefficients, patient counts and bootstrap intervals retained in the source. b, Available-case mean regional EASI divided by its theoretical maximum and multiplied by 100; cells show percentages, not patient counts. c, Mean paired regional decrease as a percentage of the same regional maximum for three separately labelled intervals; positive values indicate improvement. The abro columns represent the observed pooled-dose mixture. These heatmaps summarize magnitude; exact 95% t intervals and counts are in the source tables. d, Whole-body PP-NRS mean decreases with 95% t intervals, not region-specific itch. Regional maxima are 7.2, 14.4, 21.6 and 28.8 for head/neck, upper limbs, trunk and lower limbs. All regional analyses require complete, bounded regional records consistent with total EASI. Intervals can share patients and are not successive cumulative effects. Colour bars encode distinct quantities and are not interchangeable. No baseline-adjusted regional treatment contrast passed the FDR threshold; full separate-dose comparisons remain available.

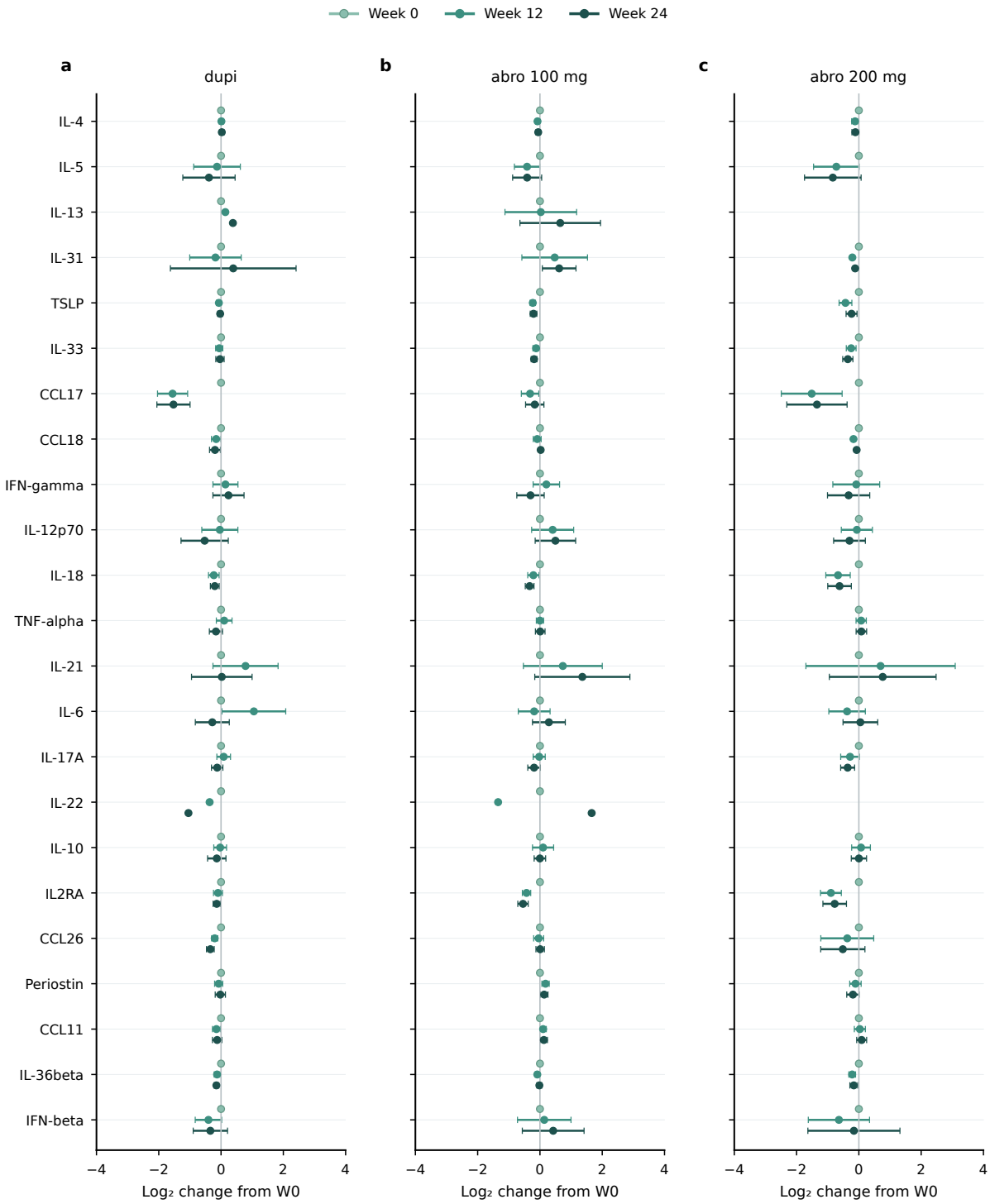


Figure S18: **Three-visit molecular trajectories in fixed patient sets.** Within each marker and treatment, the same patients contribute at W0, W12 and W24. W0 is zero by definition; W12/W24 are mean within-patient $\log_2$ changes from W0. Light blue, teal and deep blue encode W0/W12/W24 consistently across all panels. Rows are spaced to separate the three visits; horizontal intervals are 95% t confidence intervals for n$\geq$5. Smaller sets are descriptive without inferential intervals, and absent estimates remain blank. All 23 markers are retained. Fixed triplet counts are available in the source table and may differ across markers or treatments, but never across the three displayed visits within a marker/treatment.

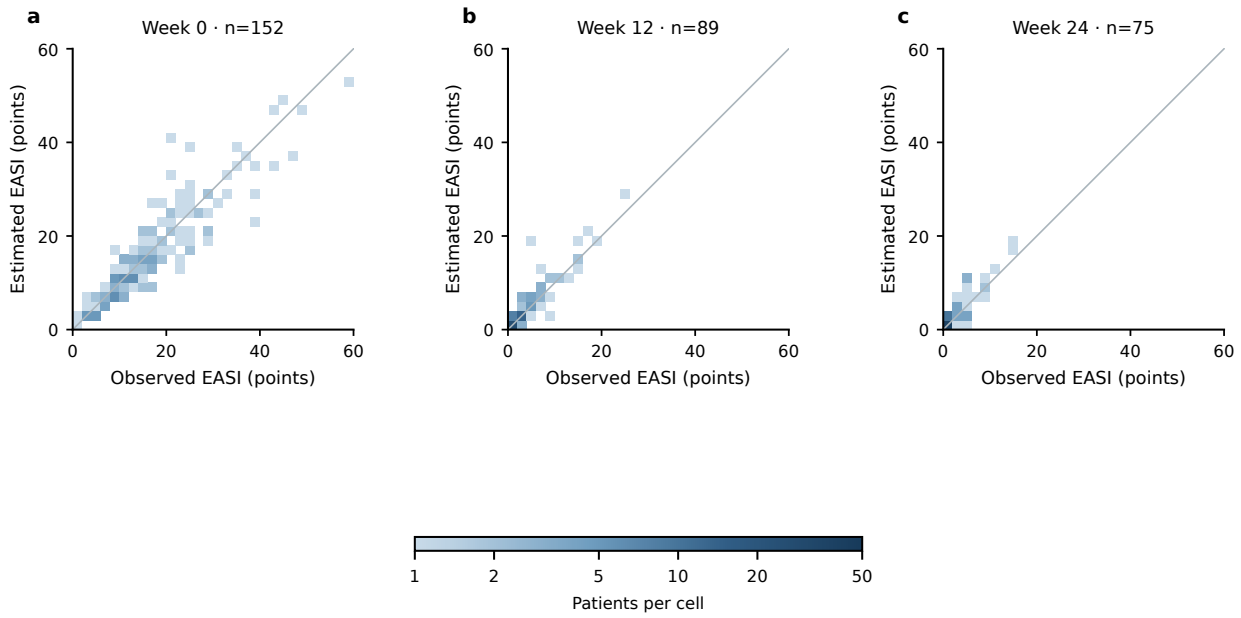


Figure S19: **Observed and estimated EASI at each visit.** a–c, Estimated versus observed EASI in the first patient-held-out repetition at W0/W12 and same-cohort temporal application at W24. Each fixed 2×2-EASI-point cell counts all exact score pairs within its boundaries. Colour gives patient count on one shared logarithmic scale; white cells contain no observations. All 152/89/75 eligible patients are retained. Identical axes preserve the severity-range comparison; grey diagonals denote exact agreement. W0/W12 use nested patient-level structure and penalty selection; earlier marker selection is not repeated. W24 had previously been inspected. These results assess the development procedure rather than independent validation of the final coefficient vector. Overall performance is in Table 2; the main figure displays the complete absolute-error distribution.

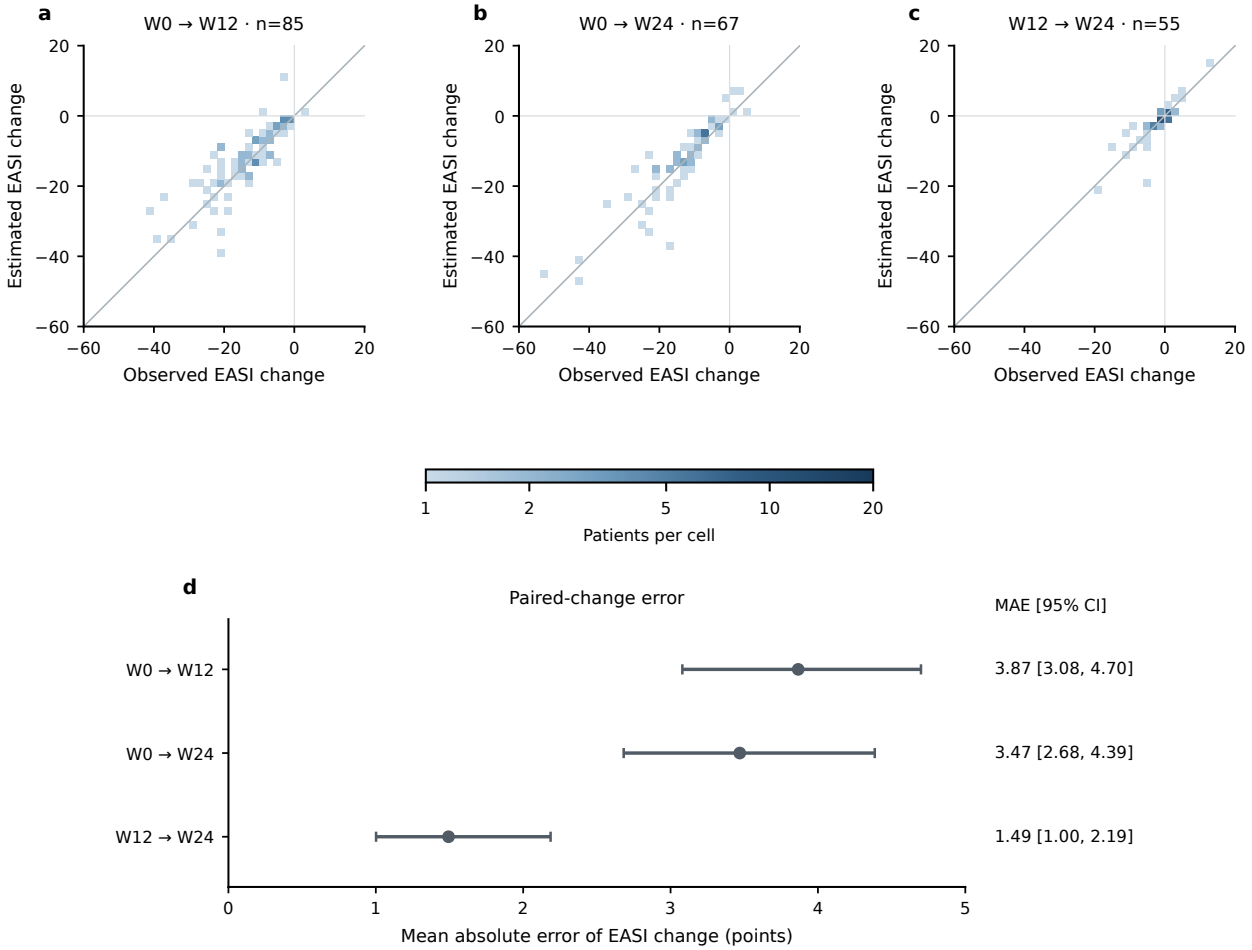


Figure S20: **Within-patient EASI change and absolute error.** a–c, Estimated versus observed EASI changes in the first patient-held-out repetition. Change is end minus start: negative values indicate improvement. Fixed 2×2-EASI-point cells count exact paired changes on one logarithmic count scale. Identical axes retain all 85, 67 and 55 pairs; the intervals can share patients and are not consecutive categories on a time axis. White cells contain no observations; diagonals mark exact agreement. d, Change MAE calculated within each of two repetitions before averaging, with 2,000 patient-bootstrap 95% confidence intervals conditional on saved predictions. Both scores are paired within each repetition before subtraction. W24 is previously inspected same-cohort temporal application. EASI75 classification is shown in the main model figure; complete EASI50/75/90 classifications are retained separately.

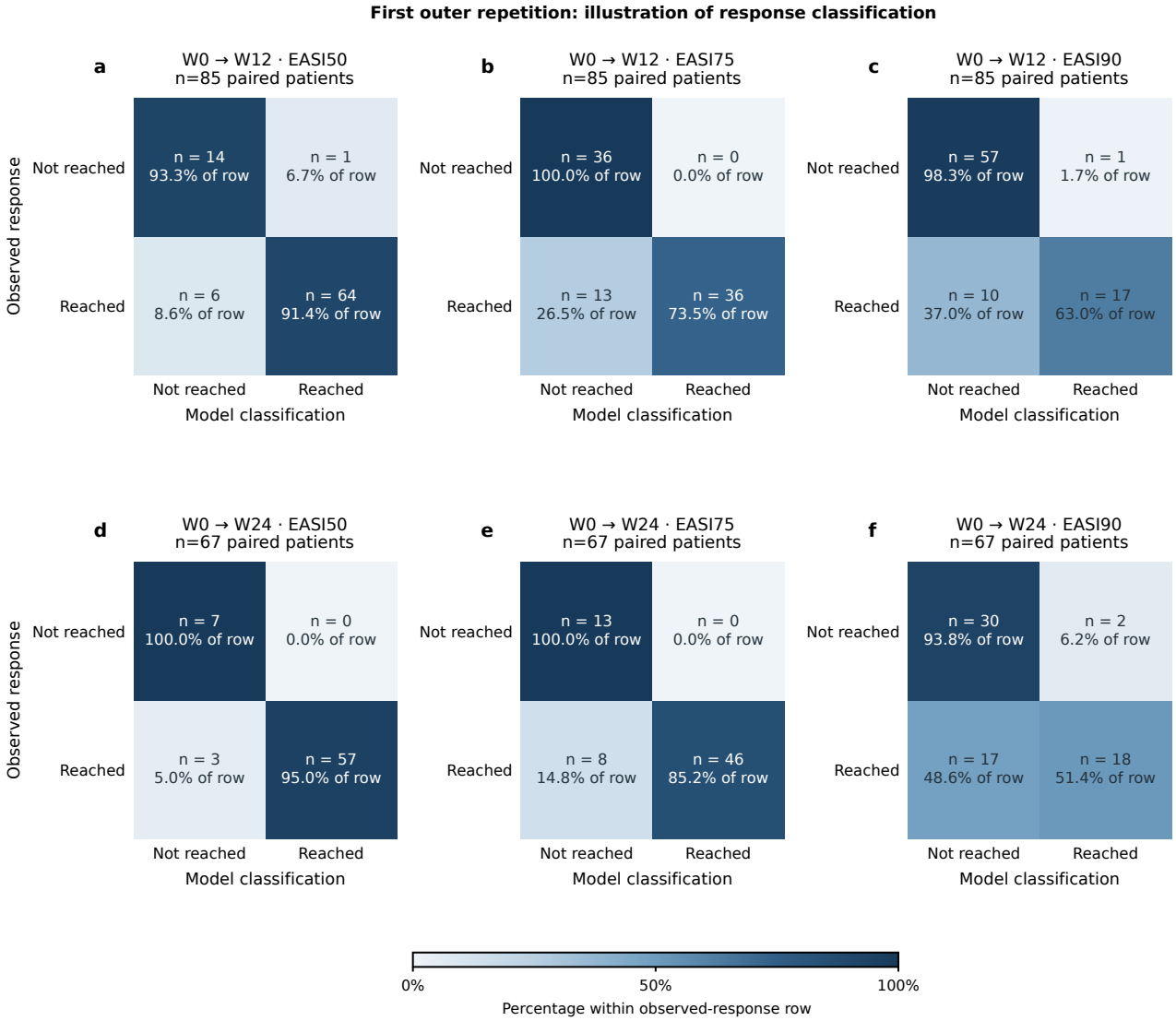


Figure S21: **Classification of EASI50, EASI75 and EASI90 response.** Each panel shows actual patient counts from the first outer repetition, separately for W0→W12 and W0→W24. Rows are observed response and columns model-estimated response. Diagonal cells are correct; upper-right cells falsely indicate response and lower-left cells miss an observed responder. Percentages are within each observed-response row, so the lower-right percentage is sensitivity and the upper-left percentage specificity. Observed response uses observed baseline/follow-up scores, whereas estimated response uses predicted baseline/follow-up scores. Counts are integers from one repetition, not the primary two-repetition average; the full metrics table retains the latter. The paired denominators are 85 and 67, with no excluded pair for invalid baseline in this displayed repetition. The two intervals can share patients. This assesses response classification, not alternative-treatment outcomes or continuous-score equivalence.